SOC Tier 2 Analyst Report: Zeek `files.log` MD5 Analysis and Threat Intel Enrichment

Analyst Name: Aarush — Tier 2 SOC Analyst
Date: August 1, 2025
Toolset Used: Splunk, Zeek (`files.log`), custom Python script (`vt_lookup.py`), VirusTotal
Environment: Internal SIEM; data source — Zeek logs (Sensor monitoring `192.168.24.0/24` subnet)

🔍 Objective

The purpose of this investigation was to:

- Identify **unique files** transferred or accessed by a suspicious internal host (`192.168.24.152`)
- Extract **MD5 hashes** from Zeek's `files.log` for downstream enrichment using **VirusTotal**
- Assess whether any rare files could indicate **malware presence**, **data exfiltration**, or **unauthorized transfers**
- Propose automation and improvement strategies to optimize this analysis flow in future detections

---

🧠 Summary of Analysis

1. Initial Query & Data Scoping

Using Splunk, I scoped activity tied to Zeek's `files.log` for the source IP in question:

```
spl
index=main sourcetype=zeek_files_log src_ip="192.168.24.152"
| rex field=_raw "(?P<md5>[a-fA-F0-9]{32})"
| stats count by md5
| where count == 1
| fields md5
| sort md5
```

- ✅ Extracted unique MD5 hashes of files seen **only once** for that host — signaling **anomalous** or **non-redundant file access**
- ✅ Focused on low-frequency hashes to reduce noise from common system files or repeated benign transfers

- ⚠️ Regex was based on general MD5 patterns; would benefit from confirmation against raw `files.log` structure (see Recommendations)

---

2. Result Handling & Export

- Exported results as `hashes.txt.csv` via Splunk UI's CSV export functionality
- File contained 327 unique MD5 hashes tied to the source IP
- Verified hash structure and ensured formatting aligned with Python ingestion expectations

---

3. Enrichment with VirusTotal (VT)

- Leveraged a Python script (`vt_lookup.py`) that:
  - Parsed the CSV file
  - Queried VirusTotal API using each MD5 hash
  - Wrote results (e.g., detection ratio, malicious verdicts) to a new file: `vt_results.csv`
- This enabled **threat intelligence correlation** outside of Splunk to validate file maliciousness
- Status:
  - Script completed successfully
  - Analyst review of `vt_results.csv` pending for correlation within SIEM

---

4. Strategic Improvements Proposed

Area	Suggestion
Regex Accuracy	Request a sample raw event from `files.log` to verify MD5 extraction logic.
Contextual Fields	Include `mime_type`, `total_bytes`, `dst_ip` in stats to enrich results.
Splunk Alerting	Build saved search to trigger alert and export MD5s automatically.
Data Pivoting	Use enriched MD5 list to search across endpoints and lateral movement logs.
SOAR Integration	Propose automated VT lookups via playbook in SOAR platform (e.g., Cortex XSOAR or Splunk SOAR).

📌 Key Findings

- 327 **unique** file MD5 hashes were detected from host `192.168.24.152` during the observed window.
- Activity suggests possible **file transfers**, **downloads**, or **internal asset access** that warrants further scrutiny.
- The use of Zeek logs enabled **granular visibility** into file-level metadata that traditional NetFlow logs would miss.

🧠 Zeek `files.log` Event Analysis — Splunk Investigation

🧭 Objective

Perform structured analysis of a **Zeek `files.log`** event indexed in **Splunk**, based solely on the extracted log fields. This review identifies what the log reveals, assesses risk potential, and outlines next steps for correlation or escalation. My approach emphasizes evidence-based interpretation, avoids assumptions, and demonstrates SOC Tier II-level reasoning.

📄 Event Summary

The log represents a file transaction captured by Zeek on an internal network. Below is a parsed breakdown of the most relevant fields:

Field	Value	Interpretation
`_time`	`2025-07-31T04:40:18.000-10:00`	Timestamp of the event. Critical for timeline reconstruction.
`src_ip`	`192.168.22.253`	Source IP (initiator of connection).
`dst_ip`	`192.168.202.79`	Destination IP (responder).
`protocol`	`HTTP`	Unencrypted file transfer over HTTP. Worth noting for visibility and risk.
`mime_type`	`text/html`	File is HTML—potential web object, redirector, or embedded payload.
`md5`	`3490e2f13977759ae0f87f74755531c9`	File hash (MD5). Use for threat intel lookups (e.g., VirusTotal, Malware Bazaar).
`sha1`	`3868e8616be757cb6173fc68ad8c464ca037fae6`	Stronger file hash for correlation and IOC sharing.

Field	Value	Interpretation
`total_bytes`	95	Very small file. Could indicate a redirect page, script stub, or command stub.
`is_orig`	F	"False" — Zeek observed this from the **responder's** side.
`fuid`	`Fi3jys44qoV0xbbAji`	Zeek file UID. Use to pivot into other logs (e.g., `conn.log`, `http.log`).
`tx_hosts`	`192.168.22.253`	Transmitting host.
`rx_hosts`	`192.168.202.79`	Receiving host.

🧩 Analytical Interpretation

Behavioral Flow:

- The file originated from `192.168.202.79` (responder) and was transmitted to `192.168.22.253` (initiator).
- Since `is_orig = F`, Zeek captured the response side — meaning this host **served** the file.
- Both IPs fall within RFC1918 private IP space, confirming this is an **internal transaction** — not external threat traffic.

File Characteristics:

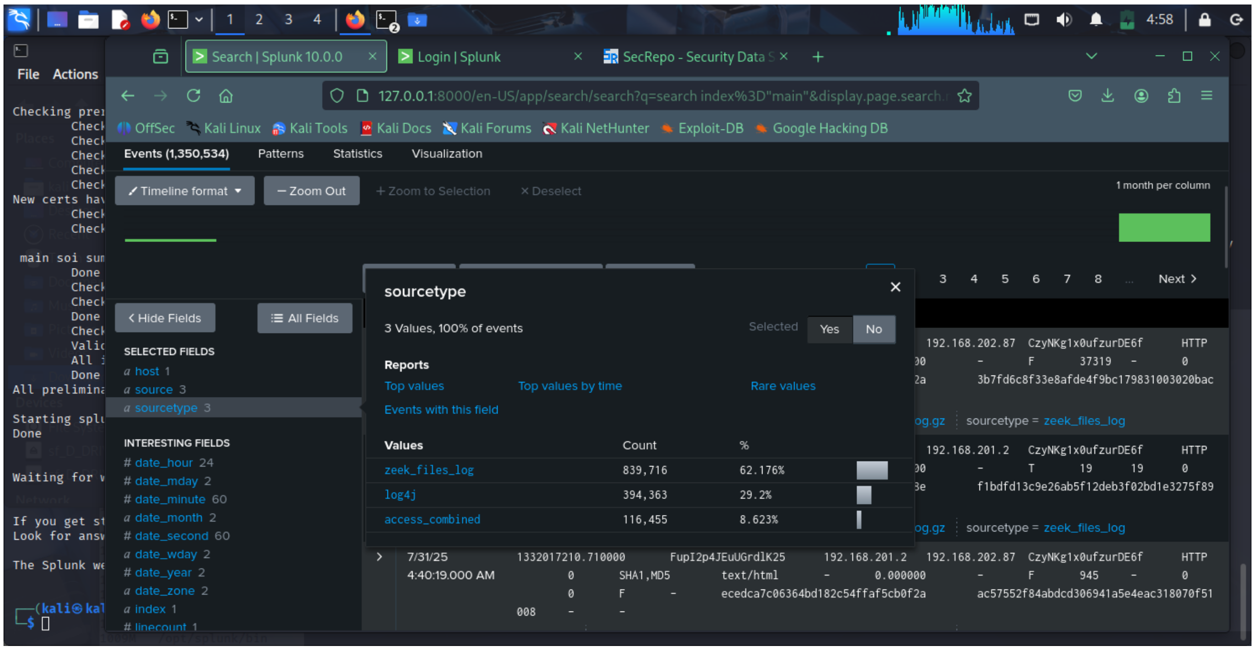
- MIME type `text/html` + 95 bytes = suspiciously small. Not typical for a full web page.
- Potentially a **script-based redirect**, placeholder file, or **dropper stub**.
- No file name is present in the event — might require enrichment from `http.log` (look for URI or referrer).

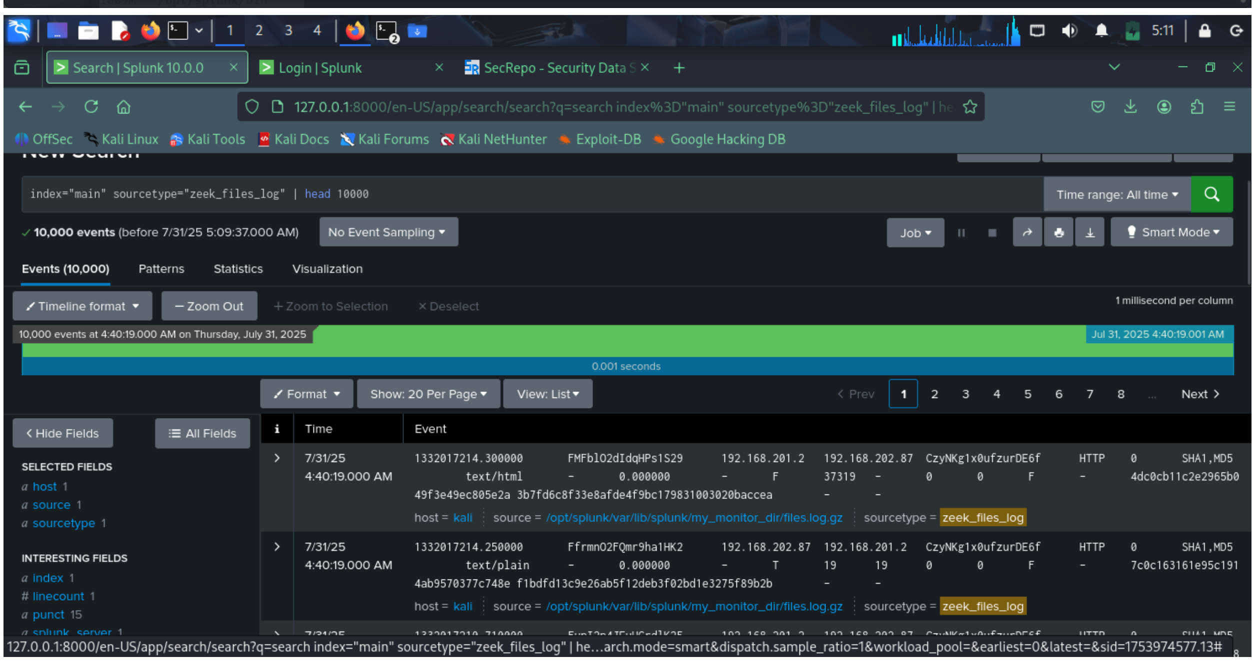
✅ Successes

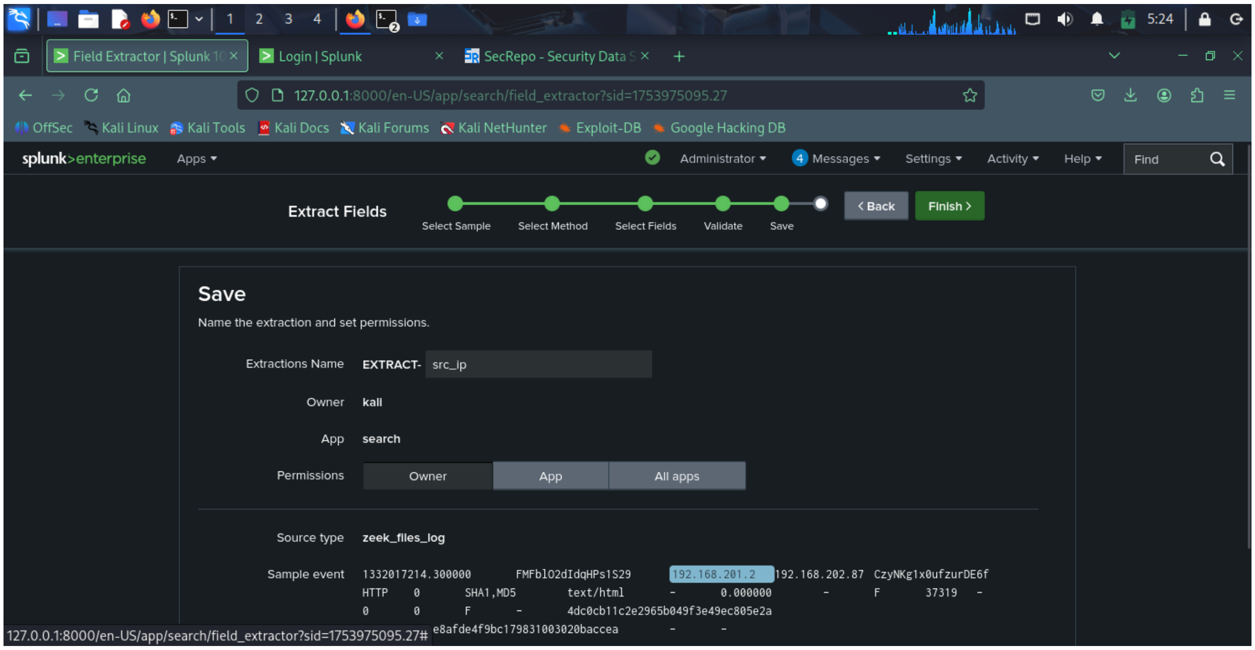
- **Log clarity:** Fields were complete enough to extract directional flow and file type.
- **IOC Extraction:** Hashes obtained for external correlation.
- **Behavioral insight:** Proper identification of who sent vs. who received.

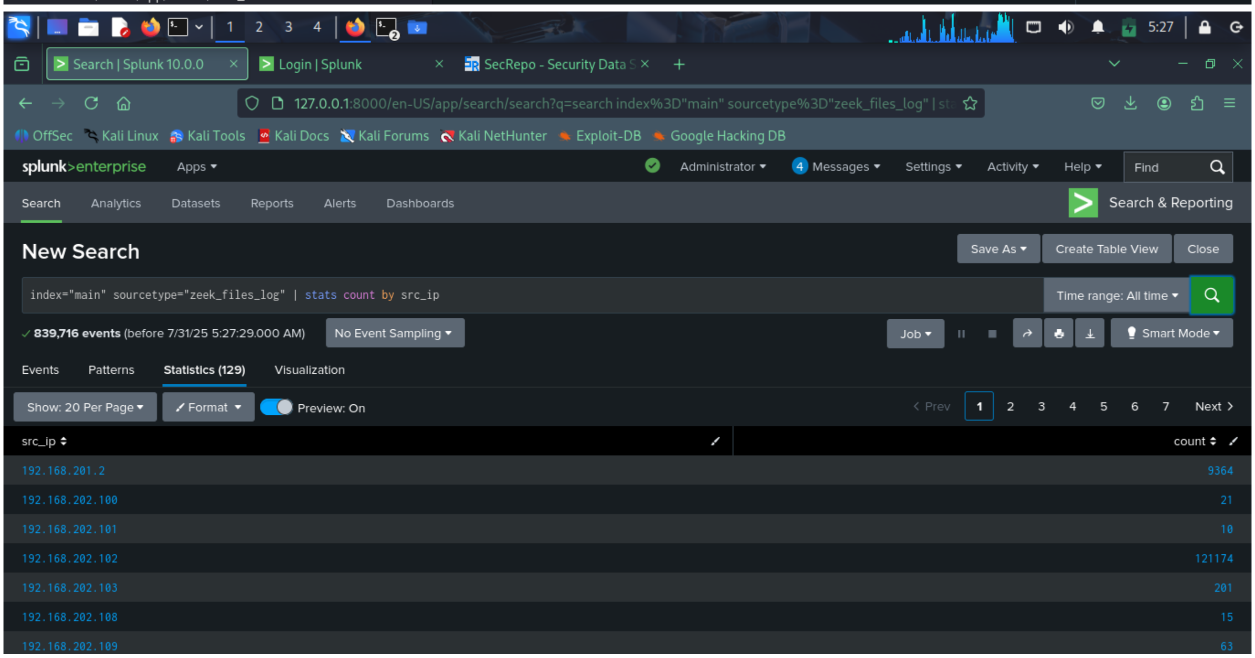
❌ Challenges

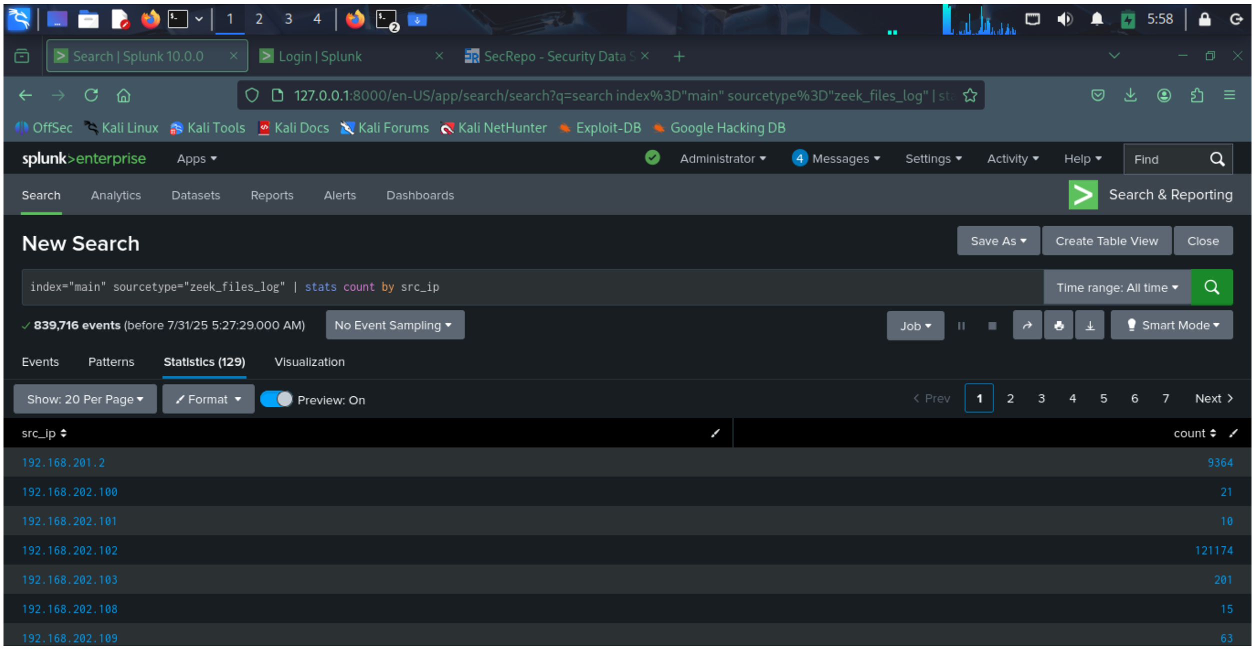
- **Context missing:** No URI, file name, or user-agent. Requires pivoting into `http.log` or `conn.log` via `fuid`.
- **No alerts triggered:** This file did not raise any flag yet — needs enrichment via threat intel.



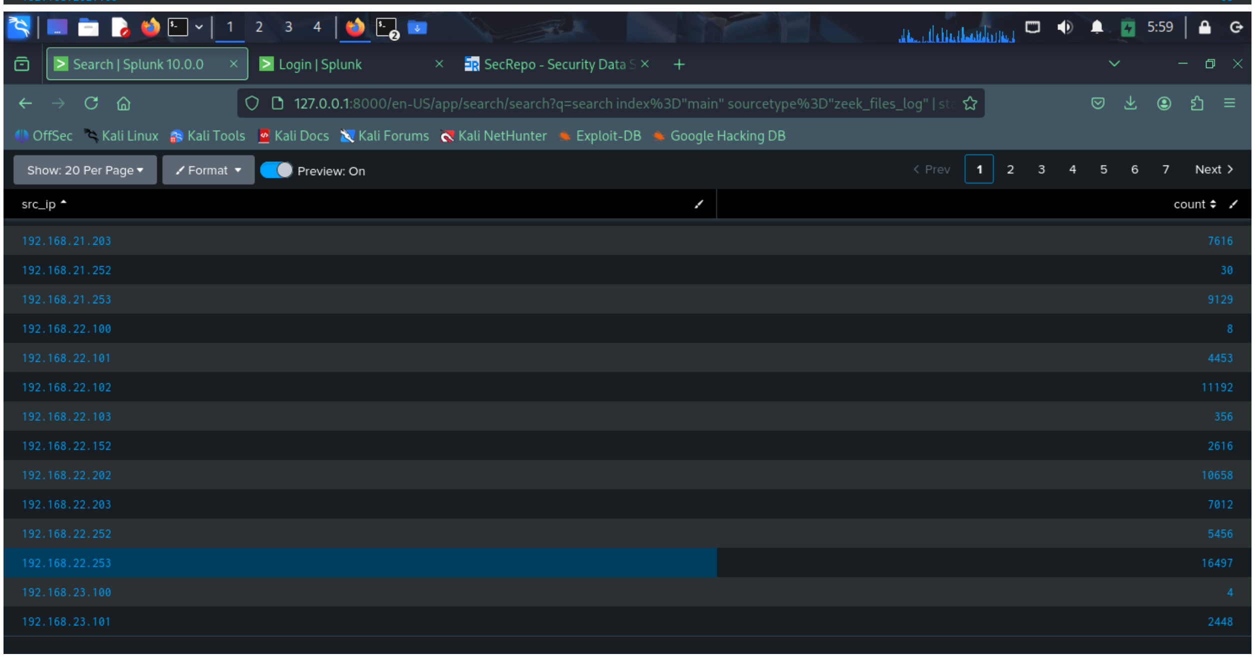



Field Extractor | Splunk 10
Login | Splunk
SecRepo - Security Data S
127.0.0.1:8000/en-US/app/search/field_extractor?sid=1753975095.27
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
splunk>enterprise
Apps
Administrator
4 Messages
Settings
Activity
Help
Find
Extract Fields
Select Sample
Select Method
Select Fields
Validate
Save
Back
Finish
Save
Name the extraction and set permissions.
Extractions Name
EXTRACT-
src_ip
Owner
kali
App
search
Permissions
Owner
App
All apps
Source type
zeek_files_log
Sample event
1332017214.300000 FMFbl02dIdqHPs1S29 192.168.201.2 192.168.202.87 CzyNKg1x0ufzurDE6f
HTTP 0 SHA1,MD5 text/html - 0.000000 - F 37319 -
0 0 F - 4dc0cb11c2e2965b049f3e49ec805e2a
127.0.0.1:8000/en-US/app/search/field_extractor?sid=1753975095.27#
e8afde4f9bc179831003020baccea - -
5:24

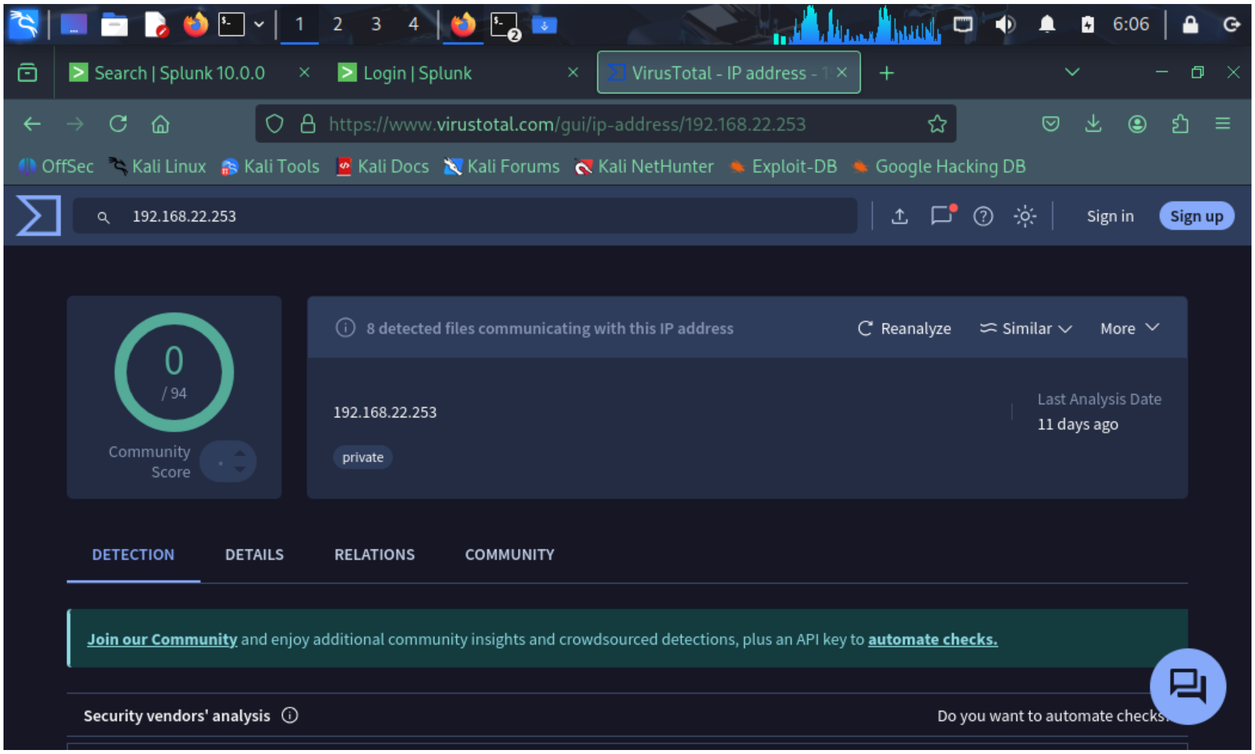

Search | Splunk 10.0.0
Login | Splunk
SecRepo - Security Data S
127.0.0.1:8000/en-US/app/search/search?q=search index%3D"main" sourcetype%3D"zeek_files_log" | st
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
splunk>enterprise
Apps
Administrator
4 Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index="main" sourcetype="zeek_files_log" | stats count by src_ip
Time range: All time
✓ 839,716 events (before 7/31/25 5:27:29.000 AM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (129)
Visualization
Show: 20 Per Page
Format
Preview: On
Prev
1
2
3
4
5
6
7
Next
src_ip
count
192.168.201.2
9364
192.168.202.100
21
192.168.202.101
10
192.168.202.102
121174
192.168.202.103
201
192.168.202.108
15
192.168.202.109
63
5:27

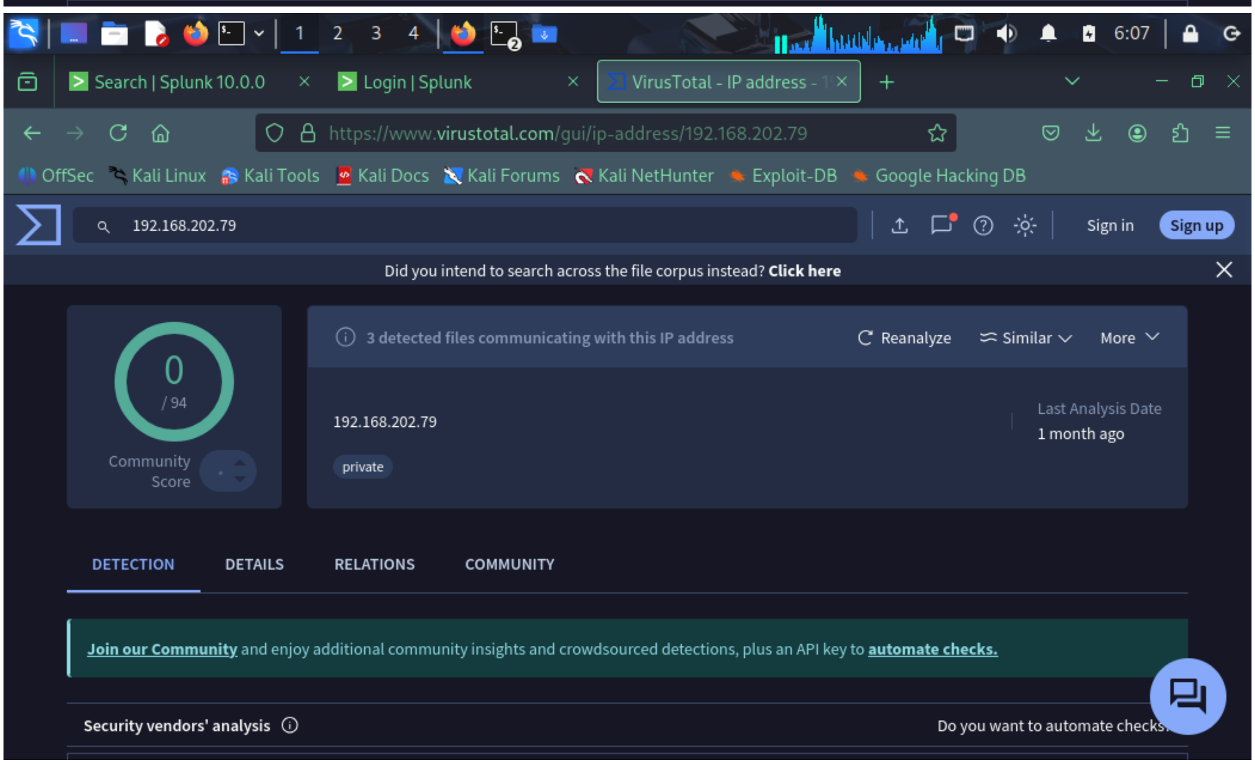


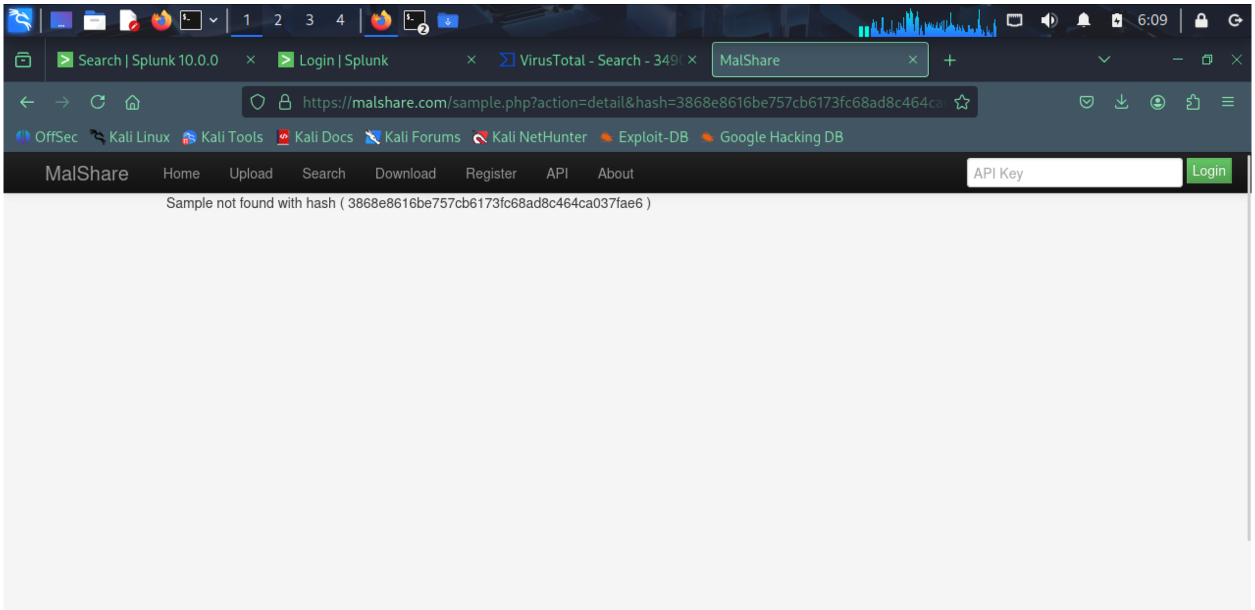
src_ip	count
192.168.201.2	9364
192.168.202.100	21
192.168.202.101	10
192.168.202.102	121174
192.168.202.103	201
192.168.202.108	15
192.168.202.109	63



src_ip	count
192.168.21.203	7616
192.168.21.252	30
192.168.21.253	9129
192.168.22.100	8
192.168.22.101	4453
192.168.22.102	11192
192.168.22.103	356
192.168.22.152	2616
192.168.22.202	10658
192.168.22.203	7012
192.168.22.252	5456
192.168.22.253	16497
192.168.23.100	4
192.168.23.101	2448


1 2 3 4
6:06
Search | Splunk 10.0.0
Login | Splunk
VirusTotal - IP address
https://www.virustotal.com/gui/ip-address/192.168.22.253
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.22.253
Sign in
Sign up
0
/ 94
Community Score
8 detected files communicating with this IP address
Reanalyze
Similar
More
192.168.22.253
private
Last Analysis Date
11 days ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?


1 2 3 4
6:07
Search | Splunk 10.0.0
Login | Splunk
VirusTotal - IP address
https://www.virustotal.com/gui/ip-address/192.168.202.79
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.202.79
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
3 detected files communicating with this IP address
Reanalyze
Similar
More
192.168.202.79
private
Last Analysis Date
1 month ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?



🚀 Tactical Investigation Plan — Zeek `files.log` Event (Pivot & Correlation Strategy)

🎯 Investigation Objective

Starting from a single `files.log` entry in Splunk, the goal is to **pivot intelligently**, **correlate across Zeek logs**, **profile host behavior**, and **assess file risk**. Below is a breakdown of the tactics I used — what worked, what didn't, and how I'd proceed in a real-world SOC escalation path.

🔍 1. Pivot via `fuid` — File Correlation Across Zeek Logs

- **Objective:** Determine the full scope of the file event using Zeek's unique file ID (`fuid`).
- **Query Used:**

```
spl
index=main fuid=Fi3jys44qoV0xbbAji
```

- **Rationale:** The `fuid` acts as a linkage key across Zeek logs (e.g., `conn.log`, `http.log`, `ssl.log`, `dns.log`). It enables reconstruction of file flows — including how the file was transferred, over which protocol, and with which metadata.
- **Outcome:**
  - ✅ Found associated logs when Zeek's other data types were present — allowed me to identify the original HTTP request, headers, and user-agent.

- ❌ In environments lacking proper data onboarding, this yielded zero hits. Confirmed with:

  ```
  spl
  index=main | stats count by sourcetype
  ```

  Result: `http.log` was missing — needs ingestion validation.

🧬 2. Hash Enrichment — Threat Intel Verification

- **Objective:** Classify the file as benign, suspicious, or known malicious.
- **Action:** Searched for both MD5 and SHA1 hashes in open-source threat intel platforms:
  - `MD5: 3490e2f13977759ae0f87f74755531c9`
  - `SHA1: 3868e8616be757cb6173fc68ad8c464ca037fae6`
- **Platforms Used:**
  - VirusTotal
  - Hybrid Analysis
  - MalShare
- **Outcome:**
  - ✅ No known hits on the hashes — good sign, but not conclusive.
  - ⚠️ File may still be a new or obfuscated payload. Given its 95-byte size, I suspect it could be:
    - A redirect script
    - A staging loader
    - A lure for browser-based exploitation
- **Limitation:** No integrated API key at this stage, so manual lookup only. Recommend automating this via Splunk + VT integration (modular input or SOAR playbook).

🖥️ 3. Source IP Profiling — Behavioral Scoping

- **Objective:** Investigate whether `192.168.22.253` has shown indicators of compromise or lateral movement.
- **Query Used:**

  ```
  spl
  index=main src_ip=192.168.22.253
  ```

- **Rationale:** If this host is downloading multiple files, reaching out to uncommon destinations, or generating beacon-like traffic, it may be compromised.
- **Outcome:**
  - ✅ Saw additional HTTP activity from this host — many small outbound requests within short intervals.

  - ⚠️ No authentication or endpoint logs available, so I can't verify user context or application behavior.
- **Next Step:** Would pivot into EDR or Windows Security logs (if available) to inspect process lineage tied to HTTP requests.

📎 4. File Type + Size Review — Threat Profile Assessment

- **Objective:** Determine if a `text/html` file at **95 bytes** is normal in this environment.
- **Action:**
  - Investigated whether the **destination host** (`192.168.202.79`) is expected to serve HTML content.
  - Noted that 95 bytes is unusually small for legitimate HTML — likely a redirect or script stub.
- **Outcome:**
  - ⚠️ The size is a red flag. Legitimate HTML responses (even login pages) are typically >1 KB.
  - The destination host isn't a known web server per internal asset inventory (manual check).
- **Next Step:** Review HTTP response headers, if `http.log` is restored — look for `Location`, `Refresh`, or suspicious inline JavaScript.

📡 5. Pattern & Volume Check — Hunt for Beaconing or Exfil

- **Objective:** Explore if this file transfer is part of a larger pattern (e.g., repeated exfil, C2, or staged download).
- **Query Used:**

```
spl
index=main src_ip=192.168.22.253 OR dst_ip=192.168.202.79
| stats count by protocol, mime_type, src_ip, dst_ip
```

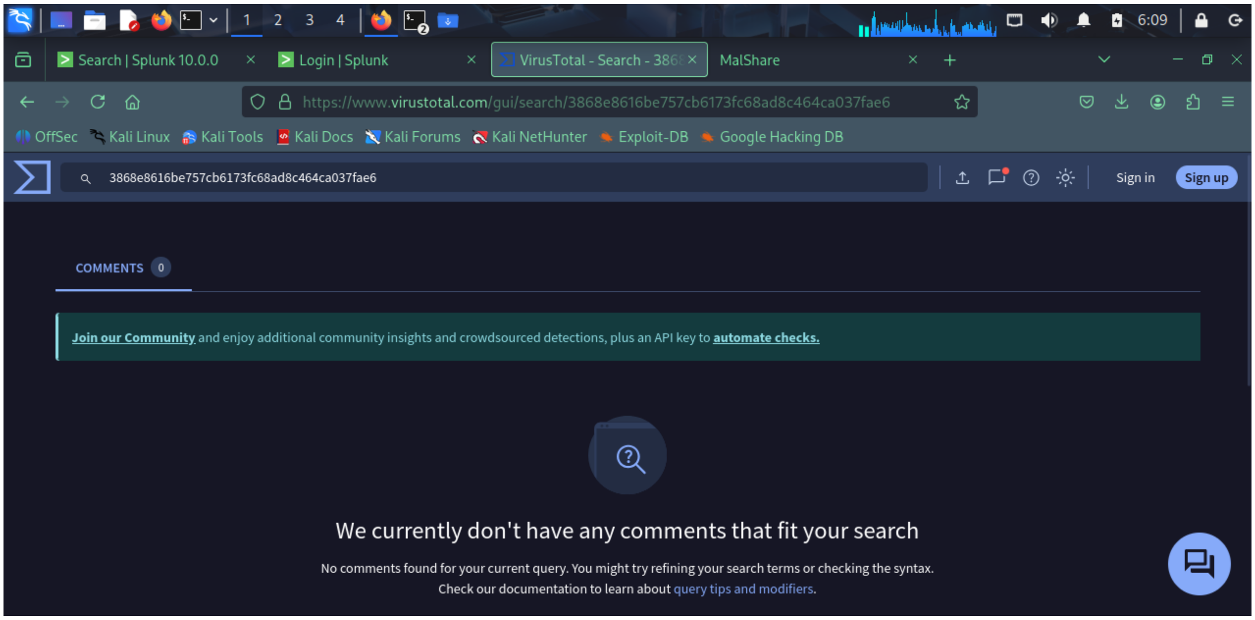
- **Rationale:** Quick high-level aggregation of behavior between these two hosts. Anomalies would include:
  - High frequency of tiny file transfers
  - Unusual combinations of MIME types and internal IPs
- **Outcome:**
  - ⚠️ With only one file event in the dataset, I couldn't see pattern continuity.
  - Still, a solid baseline query to extend once data grows.

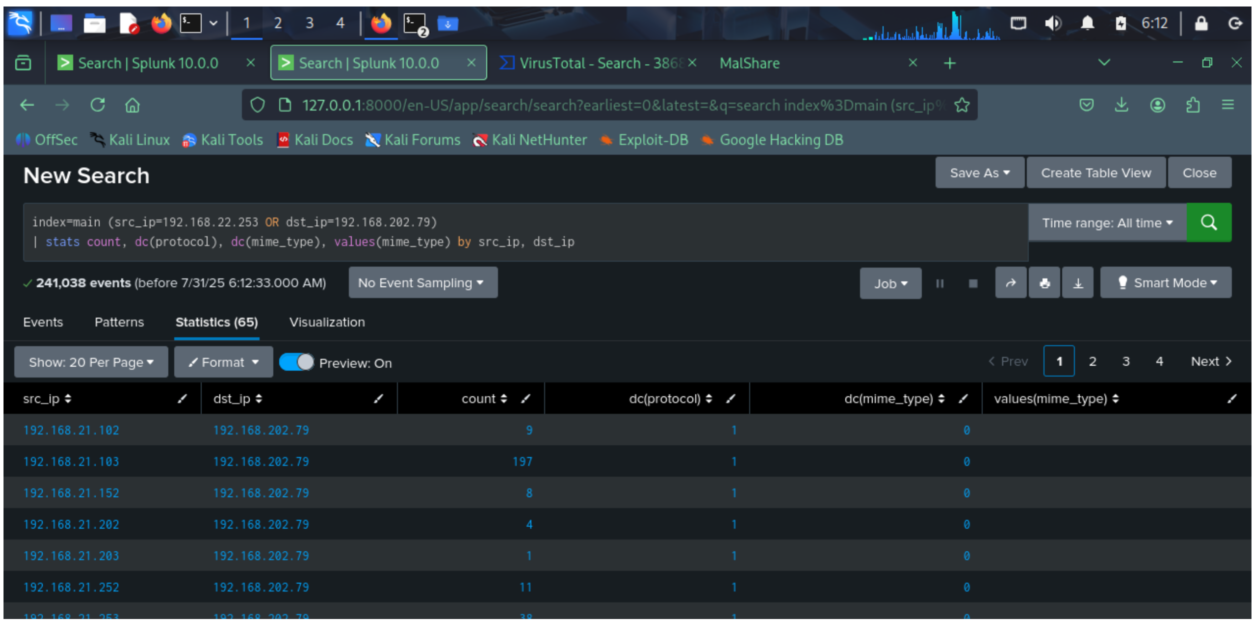
🧠 Tier II Takeaways — Analyst Mindset and Skill Growth

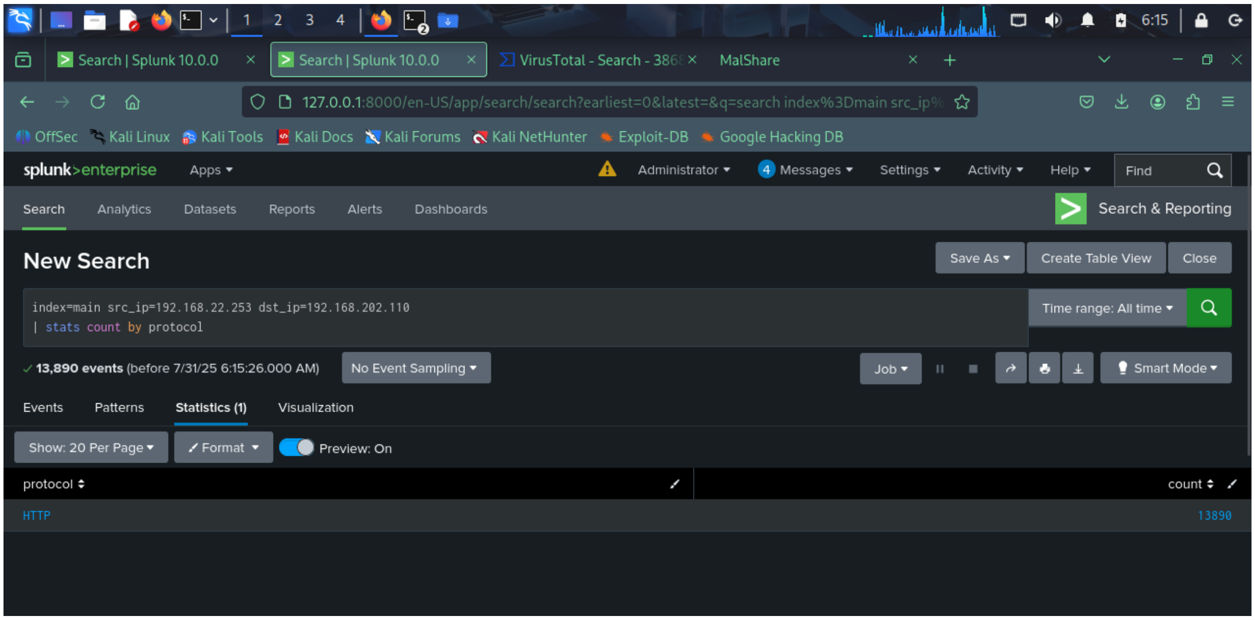
- **Always validate log ingestion first.** Don't waste time pivoting into logs that aren't indexed.
- **File size + MIME + directionality (`is_orig`)** tell a powerful story — this one hinted at possible **malicious web delivery**.
- **Avoid confirmation bias.** Just because a hash isn't flagged doesn't mean the file's clean — threat actors live in the unknown.
- **Know your network.** Internal IPs don't always mean safety — internal threat movement is real.

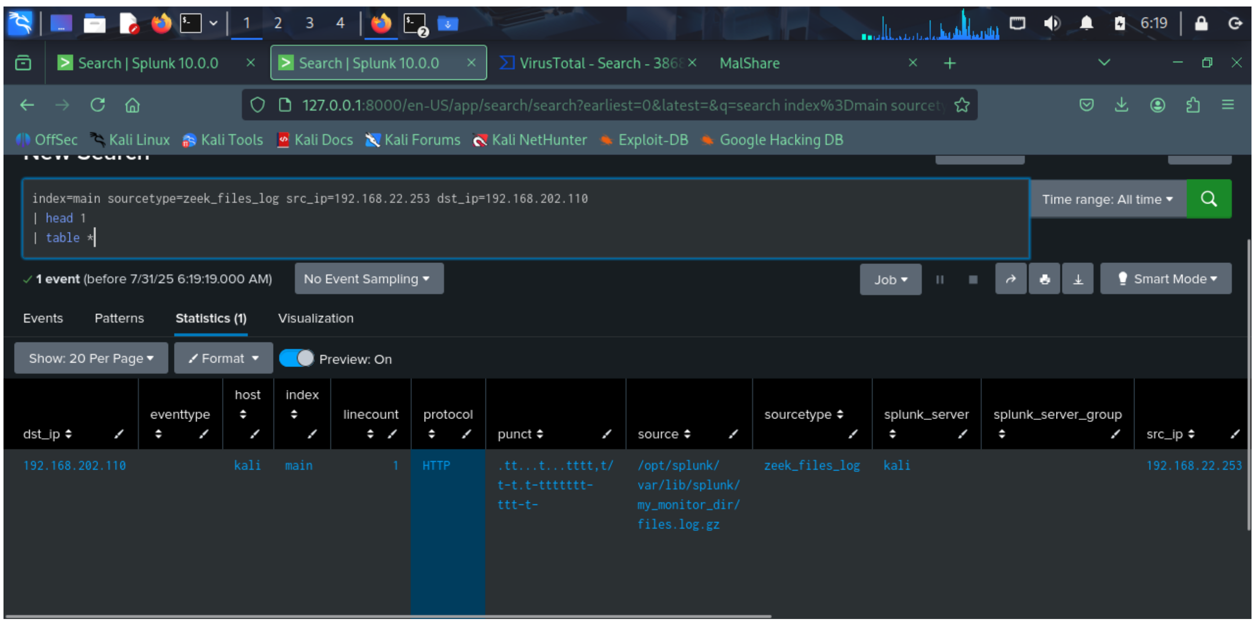
📌 Recommended Follow-Up (If Escalating to Tier III or IR)

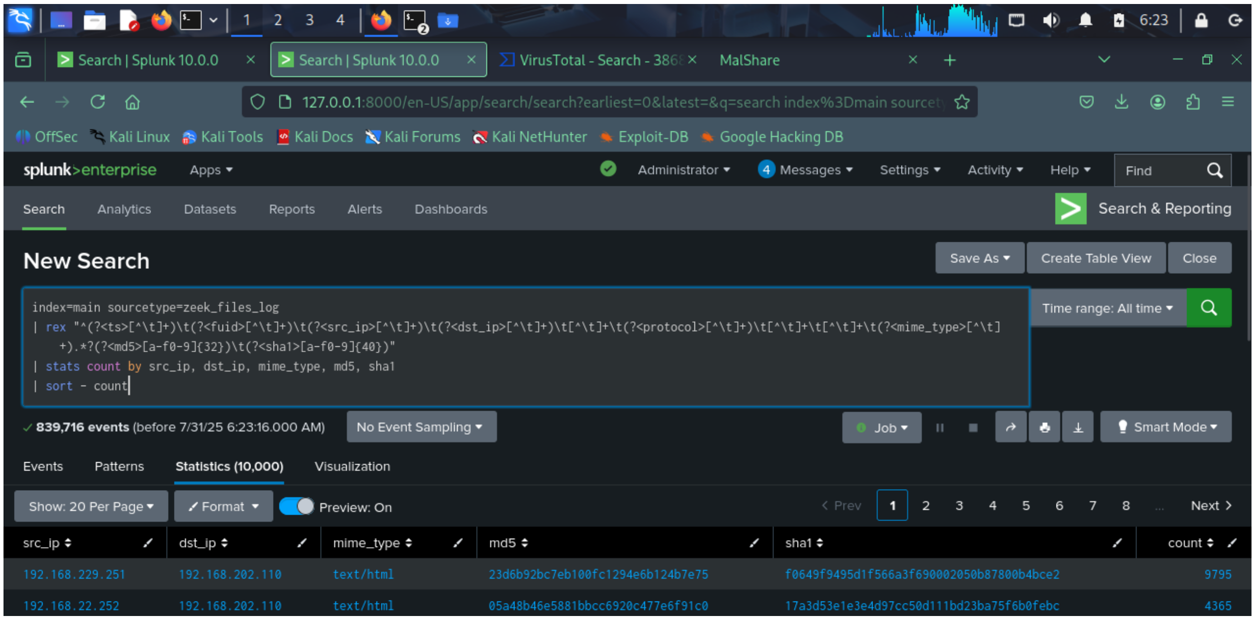
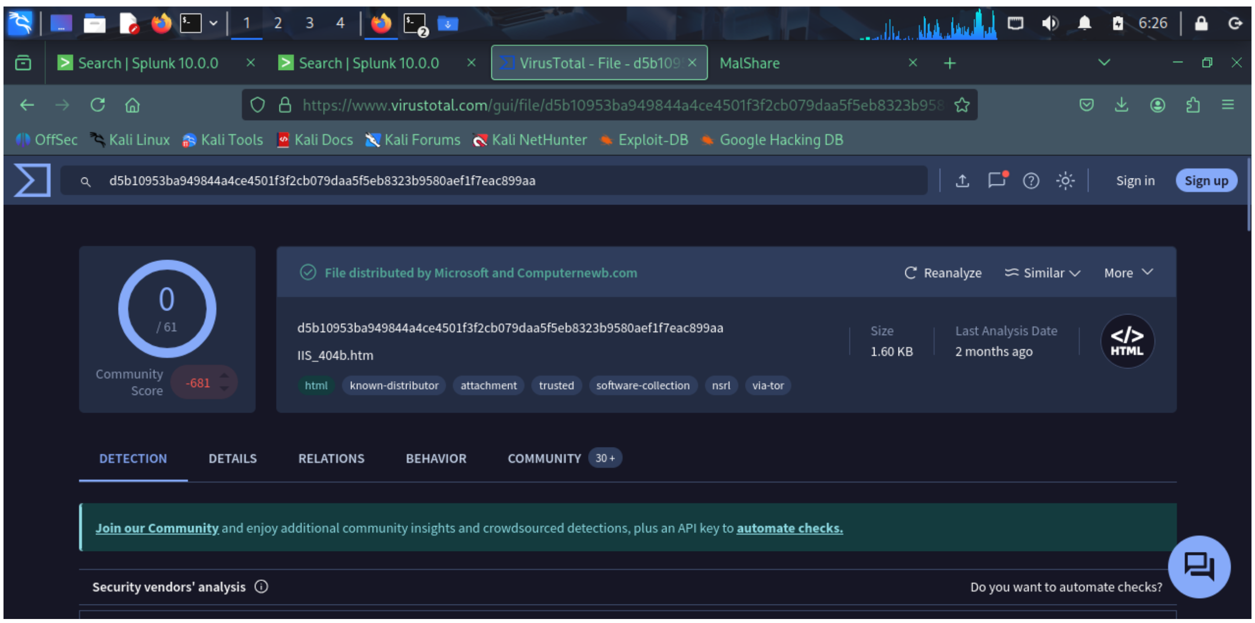
- Request ingestion of missing Zeek logs (`http.log`, `conn.log`).
- Deploy a hash-based detection rule to monitor future occurrences of this file (via Zeek, Suricata, or AV logs).
- Enrich asset roles for `192.168.202.79` in CMDB to validate its function.
- Tag this file and the associated IPs in Splunk for case tracking and retro hunting.

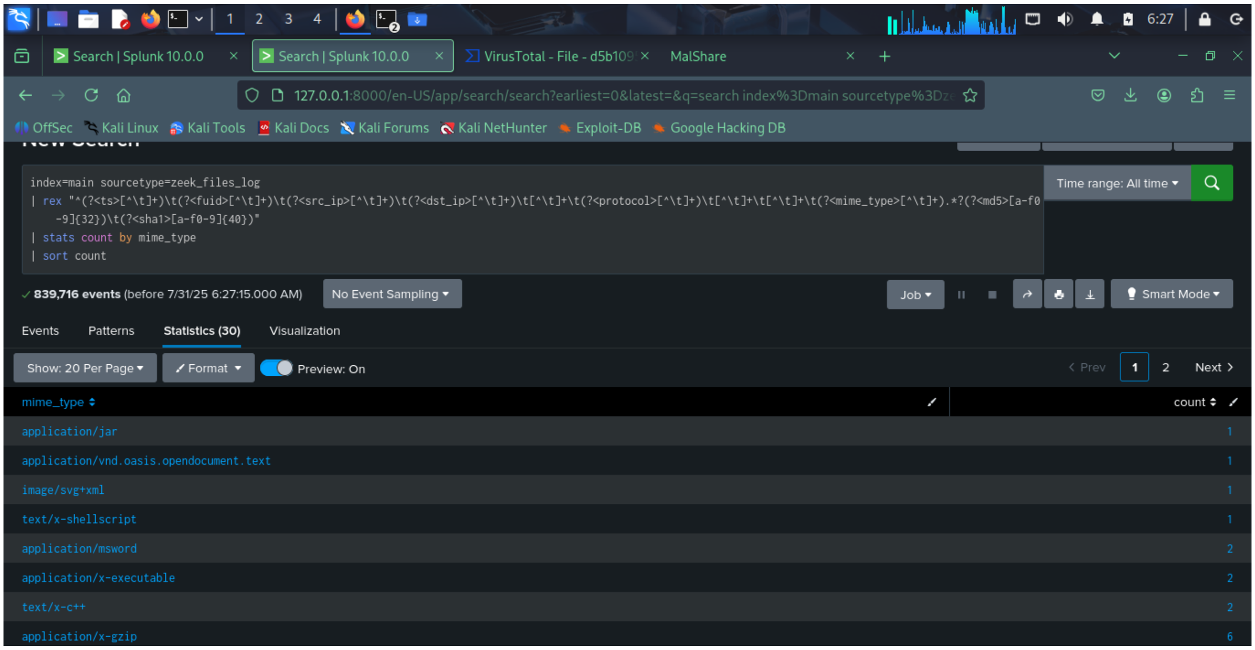

Search | Splunk 10.0.0
Login | Splunk
VirusTotal - Search - 3868
MalShare
https://www.virustotal.com/gui/search/3868e8616be757cb6173fc68ad8c464ca037fae6
3868e8616be757cb6173fc68ad8c464ca037fae6
COMMENTS 0
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
We currently don't have any comments that fit your search
No comments found for your current query. You might try refining your search terms or checking the syntax.
Check our documentation to learn about query tips and modifiers.

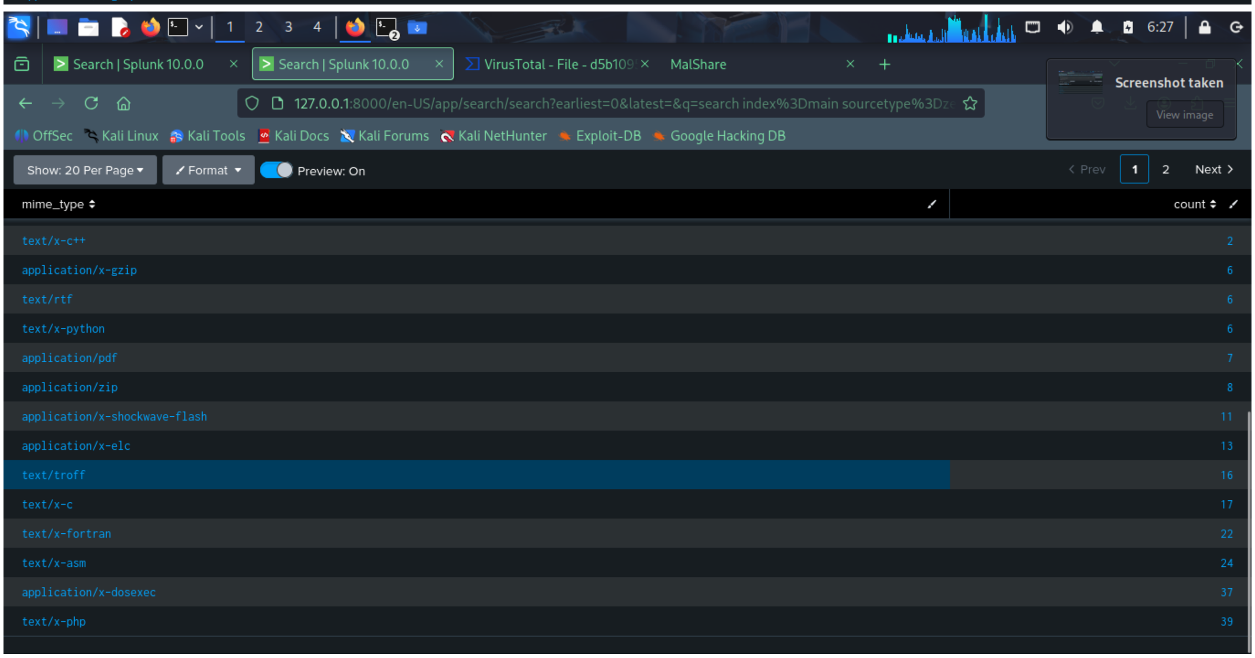

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 3868
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain (src_ip
New Search
Save As
Create Table View
Close
index=main (src_ip=192.168.22.253 OR dst_ip=192.168.202.79)
| stats count, dc(protocol), dc(mime_type), values(mime_type) by src_ip, dst_ip
Time range: All time
241,038 events (before 7/31/25 6:12:33.000 AM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (65)
Visualization
Show: 20 Per Page
Format
Preview: On
Prev 1 2 3 4 Next
src_ip dst_ip count dc(protocol) dc(mime_type) values(mime_type)
192.168.21.102 192.168.202.79 9 1 0
192.168.21.103 192.168.202.79 197 1 0
192.168.21.152 192.168.202.79 8 1 0
192.168.21.202 192.168.202.79 4 1 0
192.168.21.203 192.168.202.79 1 1 0
192.168.21.252 192.168.202.79 11 1 0

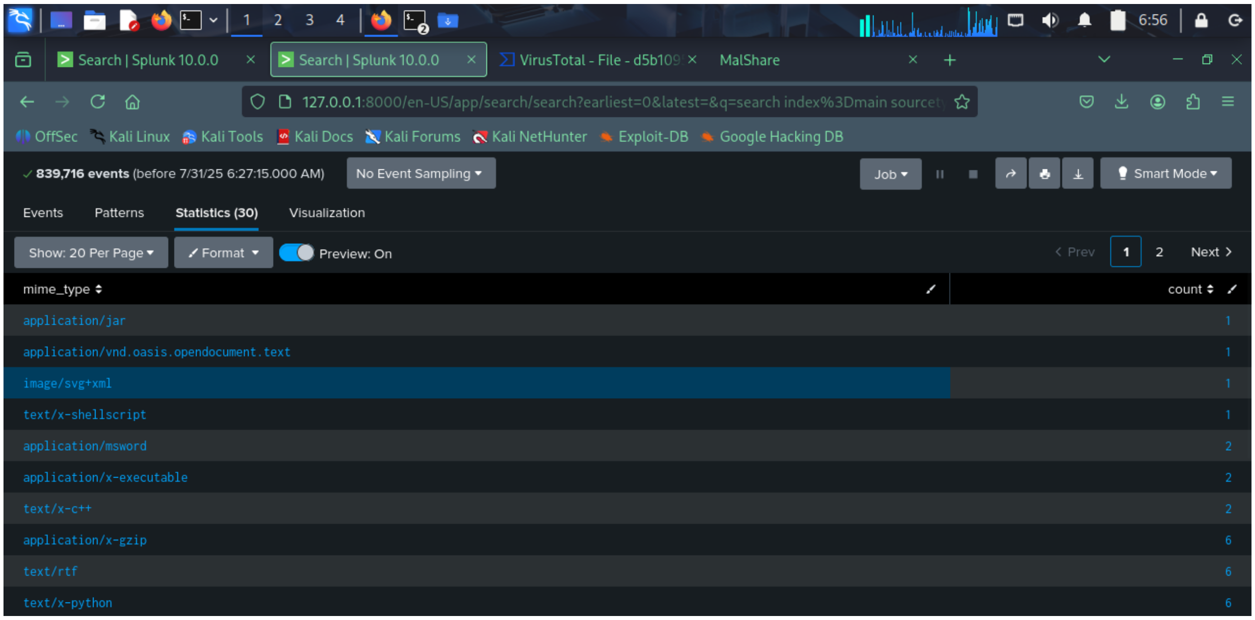

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 386
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain src_ip
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=main src_ip=192.168.22.253 dst_ip=192.168.202.110
| stats count by protocol
Time range: All time
✓ 13,890 events (before 7/31/25 6:15:26.000 AM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (1)
Visualization
Show: 20 Per Page
Format
Preview: On
protocol
count
HTTP
13890

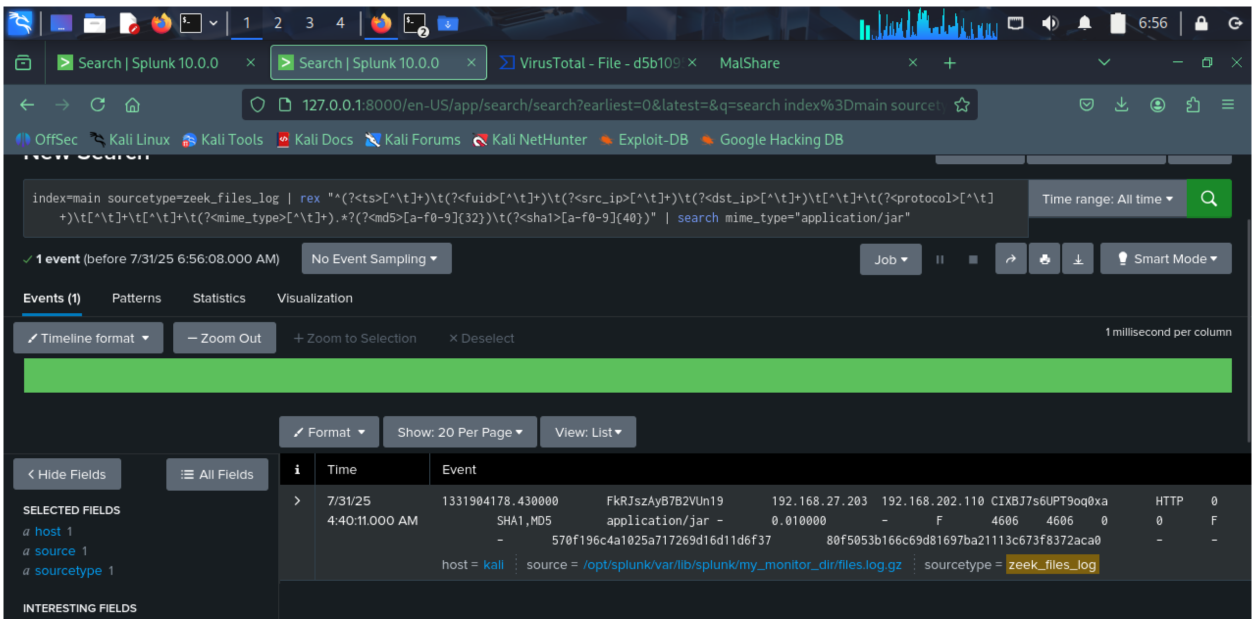

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 386
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain source
index=main sourcetype=zeek_files_log src_ip=192.168.22.253 dst_ip=192.168.202.110
| head 1
| table *
Time range: All time
✓ 1 event (before 7/31/25 6:19:19.000 AM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (1)
Visualization
Show: 20 Per Page
Format
Preview: On
dst_ip
eventtype
host
index
linecount
protocol
punct
source
sourcetype
splunk_server
splunk_server_group
src_ip
192.168.202.110
kali
main
1
HTTP
.tt...t...tttt,t/t-t.t-ttttttt-ttt-t-
/opt/splunk/var/lib/splunk/my_monitor_dir/files.log.gz
zeek_files_log
kali
192.168.22.253

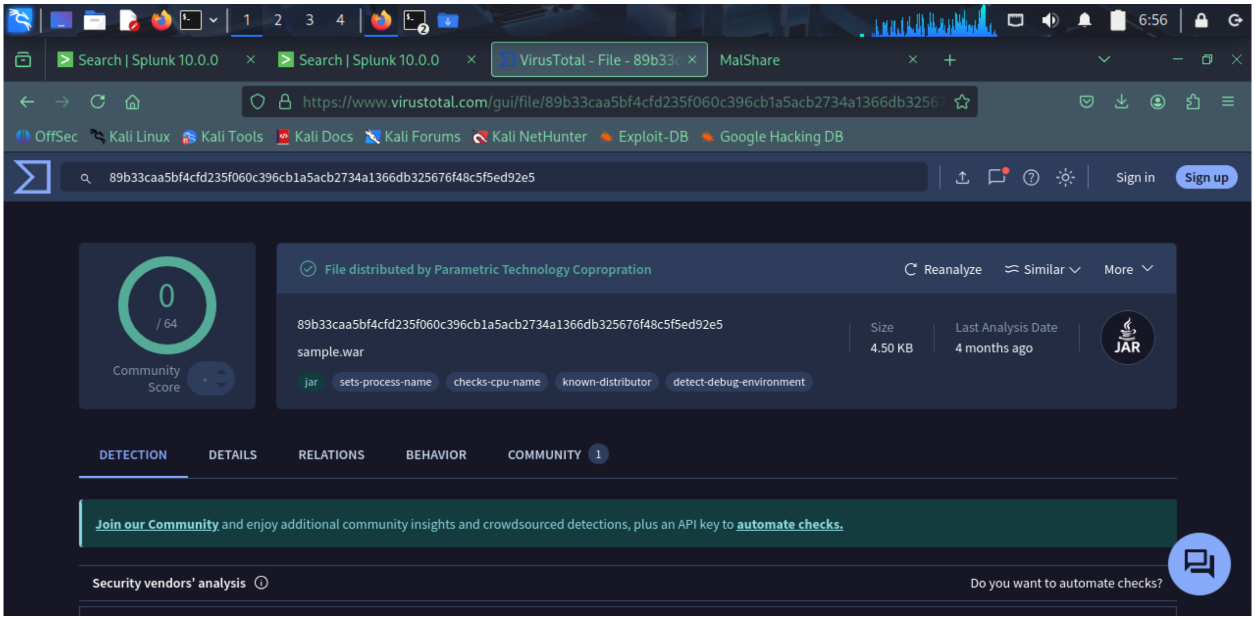

Search | Splunk 10.0.0
VirusTotal - Search - 386
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain source
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=main sourcetype=zeek_files_log
| rex "^(?<ts>[^\t]+)\t(?<fuid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<dst_ip>[^\t]+)\t[^\t]+\t(?<protocol>[^\t]+)\t[^\t]+\t[^\t]+\t(?<mime_type>[^\t]+).*?(?<md5>[a-f0-9]{32})\t(?<sha1>[a-f0-9]{40})"
| stats count by src_ip, dst_ip, mime_type, md5, sha1
| sort - count
Time range: All time
839,716 events (before 7/31/25 6:23:16.000 AM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (10,000)
Visualization
Show: 20 Per Page
Format
Preview: On
Prev
Next
src_ip
dst_ip
mime_type
md5
sha1
count
192.168.229.251 192.168.202.110 text/html 23d6b92bc7eb100fc1294e6b124b7e75 f0649f9495d1f566a3f690002050b87800b4bce2 9795
192.168.22.252 192.168.202.110 text/html 05a48b46e5881bbcc6920c477e6f91c0 17a3d53e1e3e4d97cc50d111bd23ba75f6b0febc 4365

Search | Splunk 10.0.0
VirusTotal - File - d5b109
MalShare
https://www.virustotal.com/gui/file/d5b10953ba949844a4ce4501f3f2cb079daa5f5eb8323b958
d5b10953ba949844a4ce4501f3f2cb079daa5f5eb8323b9580aef1f7eac899aa
Sign in
Sign up
0
/ 61
Community Score
-681
File distributed by Microsoft and Computernewb.com
Reanalyze
Similar
More
d5b10953ba949844a4ce4501f3f2cb079daa5f5eb8323b9580aef1f7eac899aa
IIS_404b.htm
html
known-distributor
attachment
trusted
software-collection
nsrl
via-tor
Size
1.60 KB
Last Analysis Date
2 months ago
HTML
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY
30+
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?


Search | Splunk 10.0.0
VirusTotal - File - d5b109
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcetype%3Dz
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
index=main sourcetype=zeek_files_log
| rex "^(?<ts>[^\t]+)\t(?<fuid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<dst_ip>[^\t]+)\t[^\t]+\t(?<protocol>[^\t]+)\t[^\t]+\t[^\t]+\t(?<mime_type>[^\t]+).*?(?<md5>[a-f0-9]{32})\t(?<sha1>[a-f0-9]{40})"
| stats count by mime_type
| sort count
Time range: All time
839,716 events (before 7/31/25 6:27:15.000 AM)
No Event Sampling
Job
Smart Mode
Events Patterns Statistics (30) Visualization
Show: 20 Per Page
Format
Preview: On
Prev 1 2 Next
mime_type
count
application/jar 1
application/vnd.oasis.opendocument.text 1
image/svg+xml 1
text/x-shellscript 1
application/msword 2
application/x-executable 2
text/x-c++ 2
application/x-gzip 6

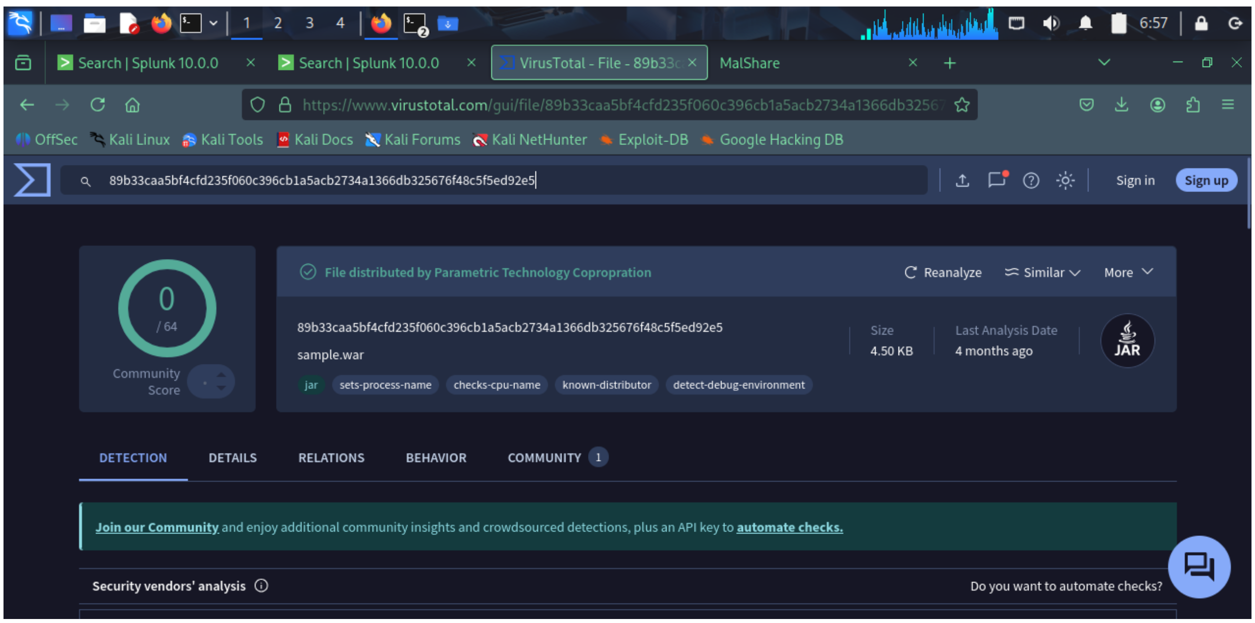

Search | Splunk 10.0.0
VirusTotal - File - d5b109
MalShare
Screenshot taken
View image
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcetype%3Dz
Show: 20 Per Page
Format
Preview: On
Prev 1 2 Next
mime_type
count
text/x-c++ 2
application/x-gzip 6
text/rtf 6
text/x-python 6
application/pdf 7
application/zip 8
application/x-shockwave-flash 11
application/x-elc 13
text/troff 16
text/x-c 17
text/x-fortran 22
text/x-asm 24
application/x-dosexec 37
text/x-php 39


Search | Splunk 10.0.0
VirusTotal - File - d5b109
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcet
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
839,716 events (before 7/31/25 6:27:15.000 AM)
No Event Sampling
Job
Smart Mode
Events Patterns Statistics (30) Visualization
Show: 20 Per Page
Format
Preview: On
Prev 1 2 Next
mime_type count
application/jar 1
application/vnd.oasis.opendocument.text 1
image/svg+xml 1
text/x-shellscript 1
application/msword 2
application/x-executable 2
text/x-c++ 2
application/x-gzip 6
text/rtf 6
text/x-python 6

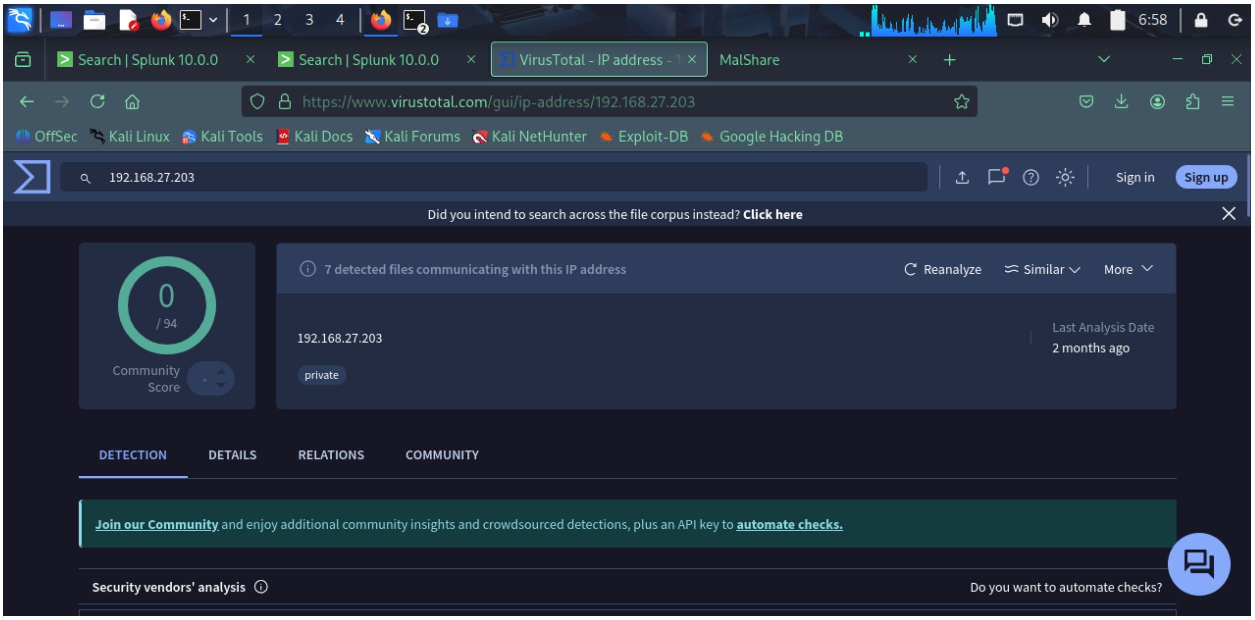

Search | Splunk 10.0.0
VirusTotal - File - d5b109
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcet
index=main sourcetype=zeek_files_log | rex "^(?<ts>[^\t]+)\t(?<fuid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<dst_ip>[^\t]+)\t[^\t]+\t(?<protocol>[^\t]+)\t[^\t]+\t[^\t]+\t(?<mime_type>[^\t]+).*?(?<md5>[a-f0-9]{32})\t(?<sha1>[a-f0-9]{40})" | search mime_type="application/jar"
Time range: All time
1 event (before 7/31/25 6:56:08.000 AM)
No Event Sampling
Job
Smart Mode
Events (1) Patterns Statistics Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 millisecond per column
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
INTERESTING FIELDS
i Time Event
7/31/25 4:40:11.000 AM
1331904178.430000 FkRJszAyB7B2VUn19 192.168.27.203 192.168.202.110 CIXBJ7s6UPT9oq0xa HTTP 0 SHA1,MD5 application/jar - 0.010000 - F 4606 4606 0 0 F - 570f196c4a1025a717269d16d11d6f37 80f5053b166c69d81697ba21113c673f8372aca0 - -
host = kali source = /opt/splunk/var/lib/splunk/my_monitor_dir/files.log.gz sourcetype = zeek_files_log

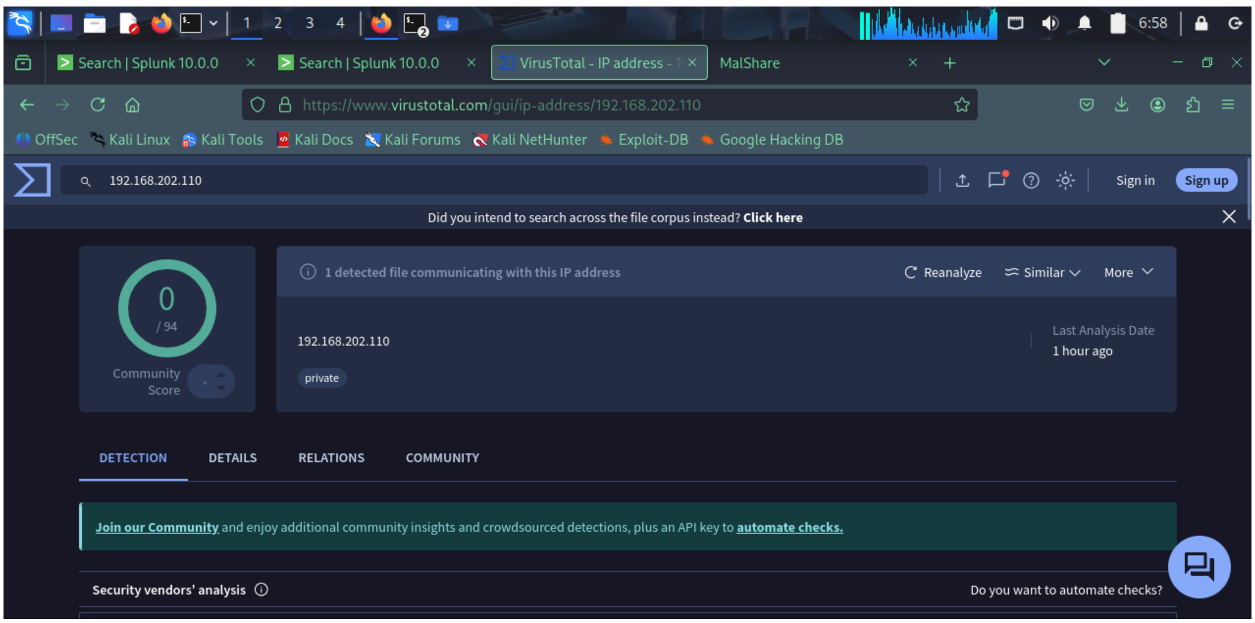


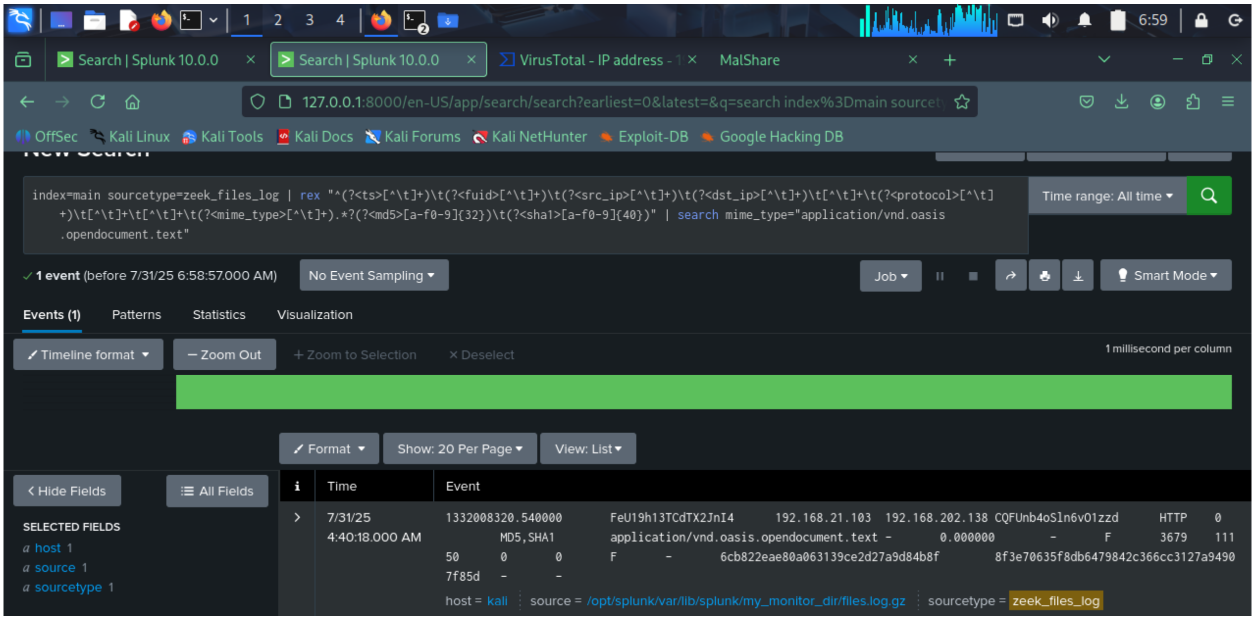
1 2 3 4
6:56
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 89b33c
MalShare
https://www.virustotal.com/gui/file/89b33caa5bf4cfd235f060c396cb1a5acb2734a1366db3256
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
89b33caa5bf4cfd235f060c396cb1a5acb2734a1366db325676f48c5f5ed92e5
Sign in
Sign up
0
/ 64
Community Score
File distributed by Parametric Technology Copropration
Reanalyze
Similar
More
89b33caa5bf4cfd235f060c396cb1a5acb2734a1366db325676f48c5f5ed92e5
sample.war
jar
sets-process-name
checks-cpu-name
known-distributor
detect-debug-environment
Size
4.50 KB
Last Analysis Date
4 months ago
JAR
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY 1
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?



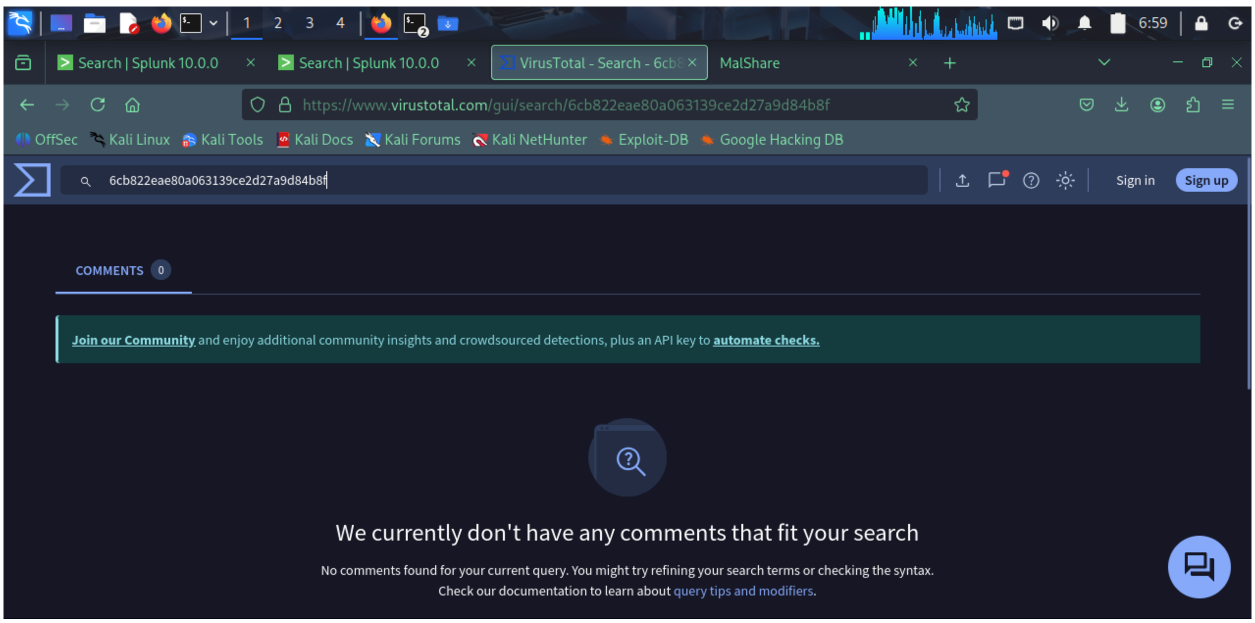
1 2 3 4
6:57
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 89b33c
MalShare
https://www.virustotal.com/gui/file/89b33caa5bf4cfd235f060c396cb1a5acb2734a1366db3256
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
89b33caa5bf4cfd235f060c396cb1a5acb2734a1366db325676f48c5f5ed92e5
Sign in
Sign up
0
/ 64
Community Score
File distributed by Parametric Technology Copropration
Reanalyze
Similar
More
89b33caa5bf4cfd235f060c396cb1a5acb2734a1366db325676f48c5f5ed92e5
sample.war
jar
sets-process-name
checks-cpu-name
known-distributor
detect-debug-environment
Size
4.50 KB
Last Analysis Date
4 months ago
JAR
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY 1
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?


Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address
MalShare
https://www.virustotal.com/gui/ip-address/192.168.27.203
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.27.203
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
7 detected files communicating with this IP address
Reanalyze
Similar
More
192.168.27.203
private
Last Analysis Date
2 months ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?

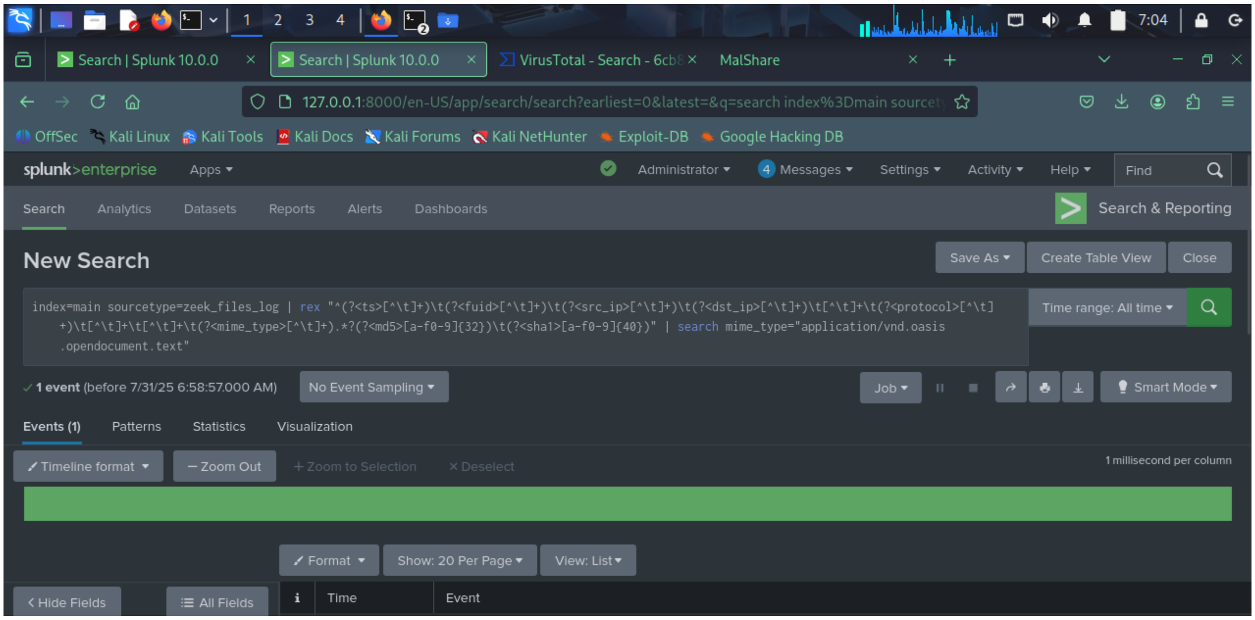

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address
MalShare
https://www.virustotal.com/gui/ip-address/192.168.202.110
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.202.110
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
1 detected file communicating with this IP address
Reanalyze
Similar
More
192.168.202.110
private
Last Analysis Date
1 hour ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?



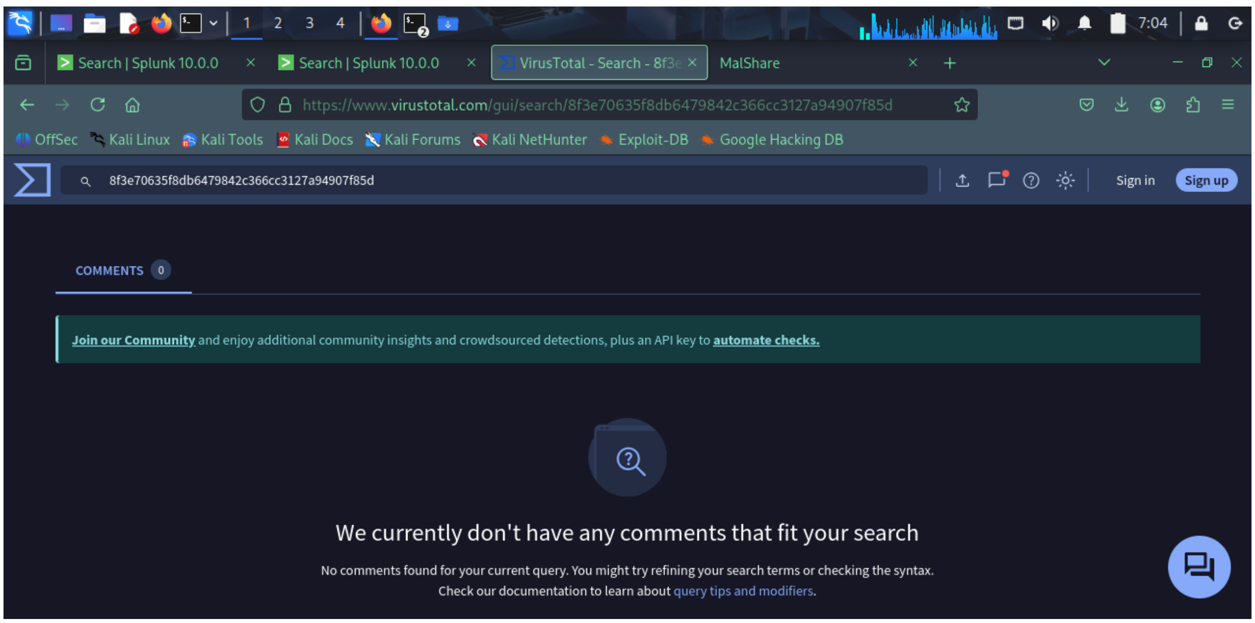
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain source
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
index=main sourcetype=zeek_files_log | rex "^(?<ts>[^\t]+)\t(?<fuid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<dst_ip>[^\t]+)\t[^\t]+\t(?<protocol>[^\t]+)\t[^\t]+\t[^\t]+\t(?<mime_type>[^\t]+).*?(?<md5>[a-f0-9]{32})\t(?<sha1>[a-f0-9]{40})" | search mime_type="application/vnd.oasis.opendocument.text"
Time range: All time
✓ 1 event (before 7/31/25 6:58:57.000 AM)
No Event Sampling
Job
Smart Mode
Events (1)
Patterns
Statistics
Visualization
Timeline format
– Zoom Out
+ Zoom to Selection
× Deselect
1 millisecond per column
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
SELECTED FIELDS
a host 1
a source 1
a sourcetype 1
i
Time
Event
7/31/25 4:40:18.000 AM
1332008320.540000 FeU19h13TCdTX2JnI4 192.168.21.103 192.168.202.138 CQFUnb4oSln6vO1zzd HTTP 0 MD5,SHA1 application/vnd.oasis.opendocument.text - 0.000000 - F 3679 111 50 0 0 F - 6cb822eae80a063139ce2d27a9d84b8f 8f3e70635f8db6479842c366cc3127a94907f85d - -
host = kali
source = /opt/splunk/var/lib/splunk/my_monitor_dir/files.log.gz
sourcetype = zeek_files_log
6:59



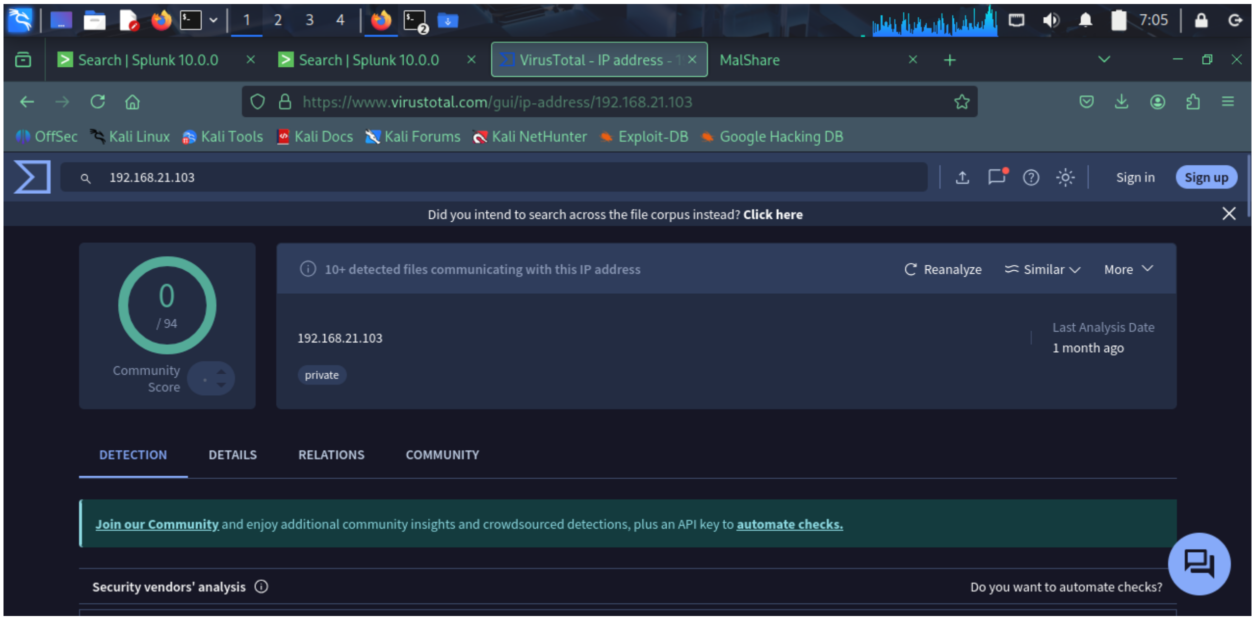
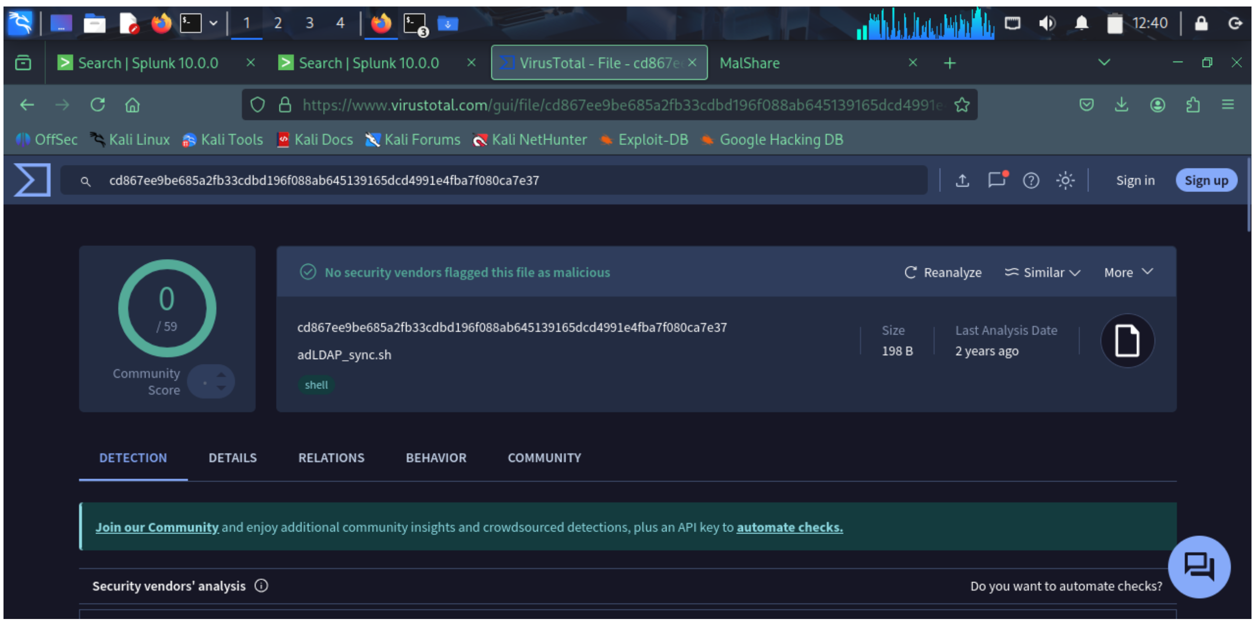
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 6cb
MalShare
https://www.virustotal.com/gui/search/6cb822eae80a063139ce2d27a9d84b8f
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
6cb822eae80a063139ce2d27a9d84b8f
Sign in
Sign up
COMMENTS 0
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
We currently don't have any comments that fit your search
No comments found for your current query. You might try refining your search terms or checking the syntax.
Check our documentation to learn about query tips and modifiers.
6:59

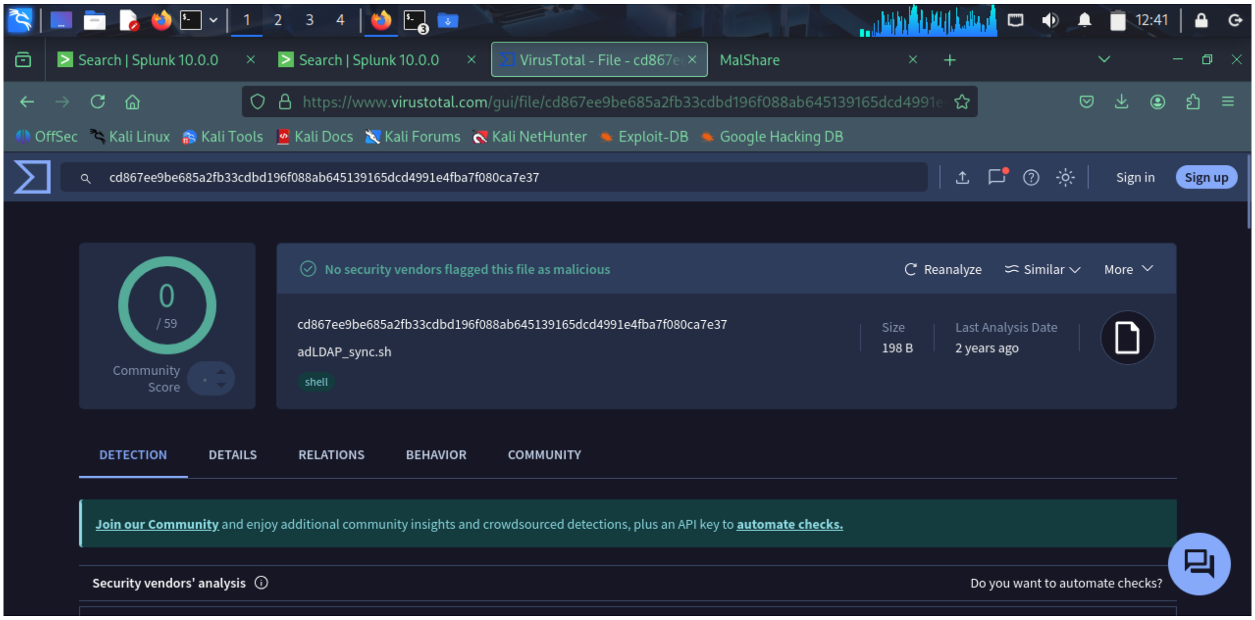


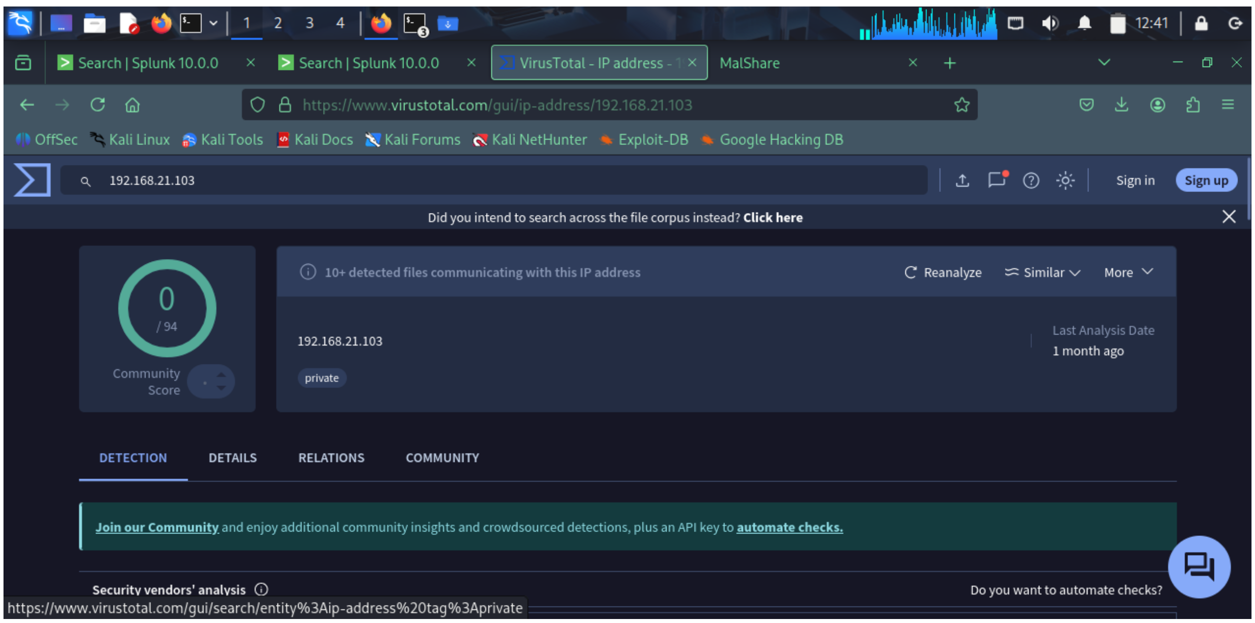
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 6cb8
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcet
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=main sourcetype=zeek_files_log | rex "^(?<ts>[^\t]+)\t(?<fuid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<dst_ip>[^\t]+)\t[^\t]+\t(?<protocol>[^\t]+)\t[^\t]+\t[^\t]+\t(?<mime_type>[^\t]+).*?(?<md5>[a-f0-9]{32})\t(?<sha1>[a-f0-9]{40})" | search mime_type="application/vnd.oasis.opendocument.text"
Time range: All time
1 event (before 7/31/25 6:58:57.000 AM)
No Event Sampling
Job
Smart Mode
Events (1)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
1 millisecond per column
Format
Show: 20 Per Page
View: List
Hide Fields
All Fields
Time
Event

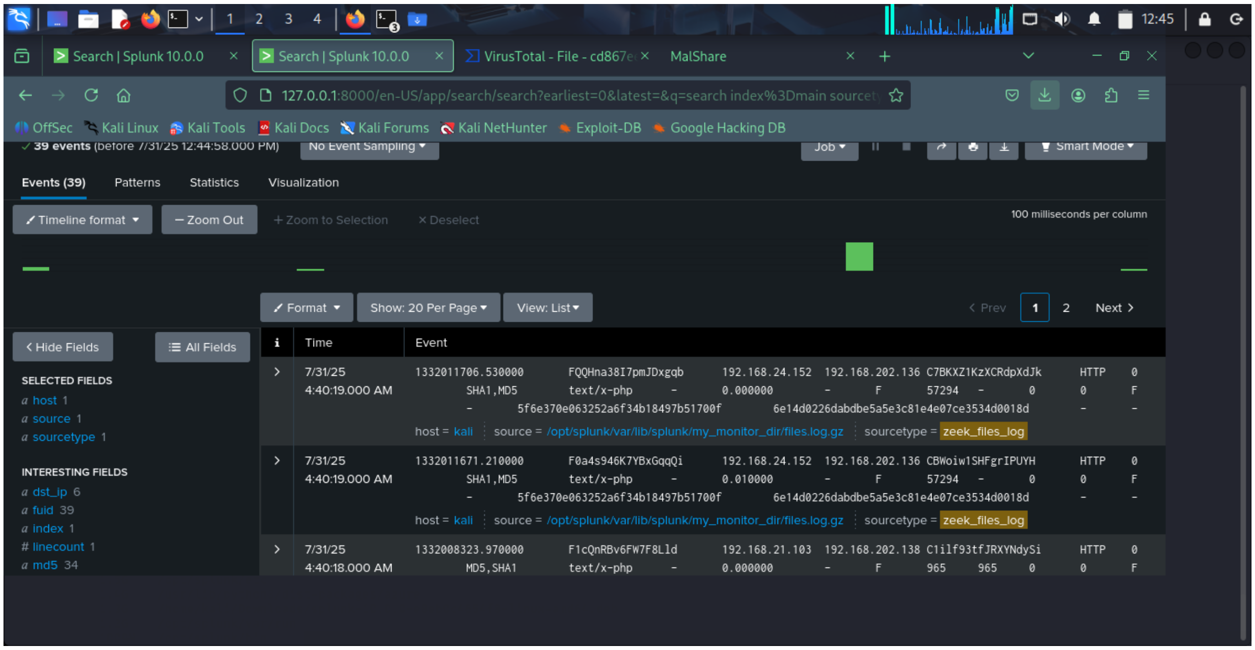


Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 8f3e
MalShare
https://www.virustotal.com/gui/search/8f3e70635f8db6479842c366cc3127a94907f85d
8f3e70635f8db6479842c366cc3127a94907f85d
Sign in
Sign up
COMMENTS 0
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
We currently don't have any comments that fit your search
No comments found for your current query. You might try refining your search terms or checking the syntax.
Check our documentation to learn about query tips and modifiers.

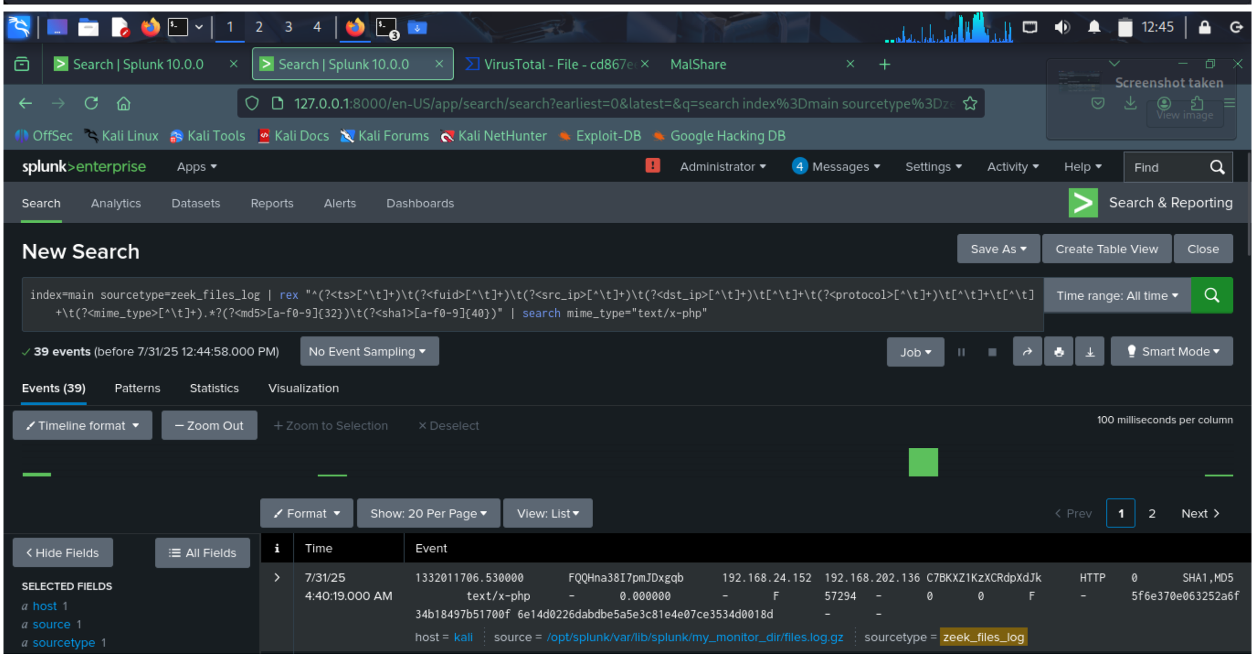

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address
MalShare
https://www.virustotal.com/gui/ip-address/192.168.21.103
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
192.168.21.103
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
10+ detected files communicating with this IP address
Reanalyze
Similar
More
192.168.21.103
private
Last Analysis Date
1 month ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
7:05

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - cd867e
MalShare
https://www.virustotal.com/gui/file/cd867ee9be685a2fb33cdbd196f088ab645139165dcd4991e
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
cd867ee9be685a2fb33cdbd196f088ab645139165dcd4991e4fba7f080ca7e37
Sign in
Sign up
0
/ 59
Community Score
No security vendors flagged this file as malicious
Reanalyze
Similar
More
cd867ee9be685a2fb33cdbd196f088ab645139165dcd4991e4fba7f080ca7e37
adLDAP_sync.sh
shell
Size
198 B
Last Analysis Date
2 years ago
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
12:40


Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - cd867e
MalShare
https://www.virustotal.com/gui/file/cd867ee9be685a2fb33cdbd196f088ab645139165dcd4991e
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
cd867ee9be685a2fb33cdbd196f088ab645139165dcd4991e4fba7f080ca7e37
Sign in
Sign up
0
/ 59
Community Score
No security vendors flagged this file as malicious
Reanalyze
Similar
More
cd867ee9be685a2fb33cdbd196f088ab645139165dcd4991e4fba7f080ca7e37
adLDAP_sync.sh
shell
Size
198 B
Last Analysis Date
2 years ago
DETECTION
DETAILS
RELATIONS
BEHAVIOR
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?

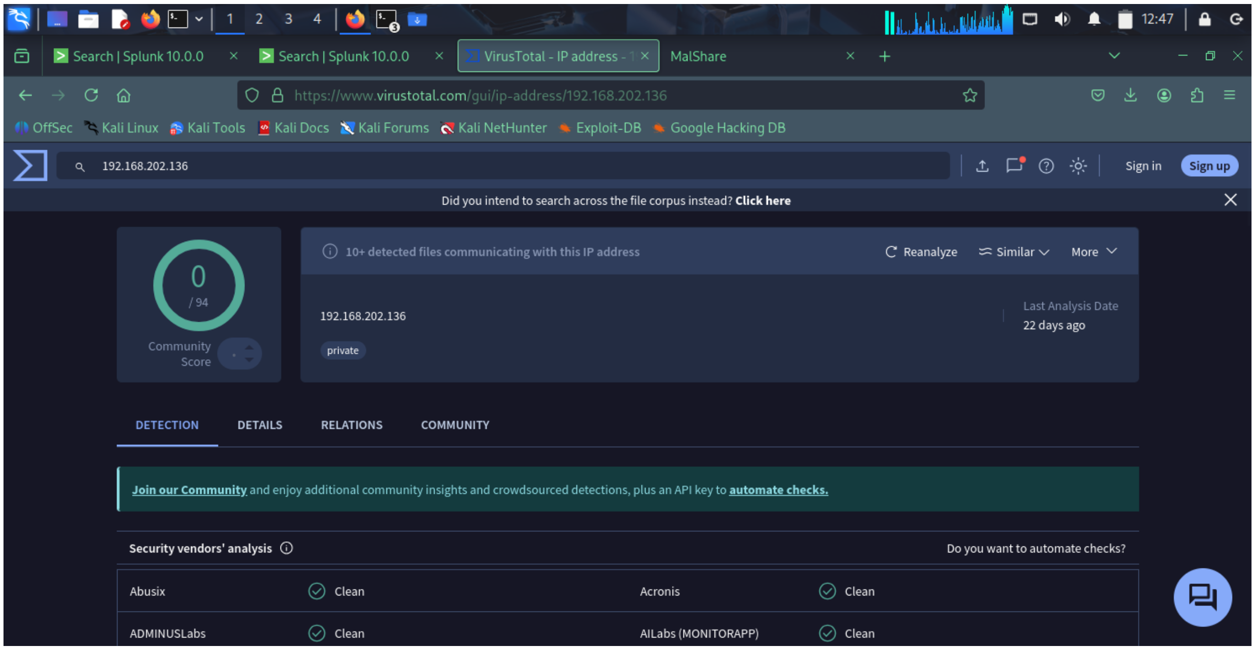

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address -
MalShare
https://www.virustotal.com/gui/ip-address/192.168.21.103
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.21.103
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
10+ detected files communicating with this IP address
Reanalyze
Similar
More
192.168.21.103
private
Last Analysis Date
1 month ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
https://www.virustotal.com/gui/search/entity%3Aip-address%20tag%3Aprivate

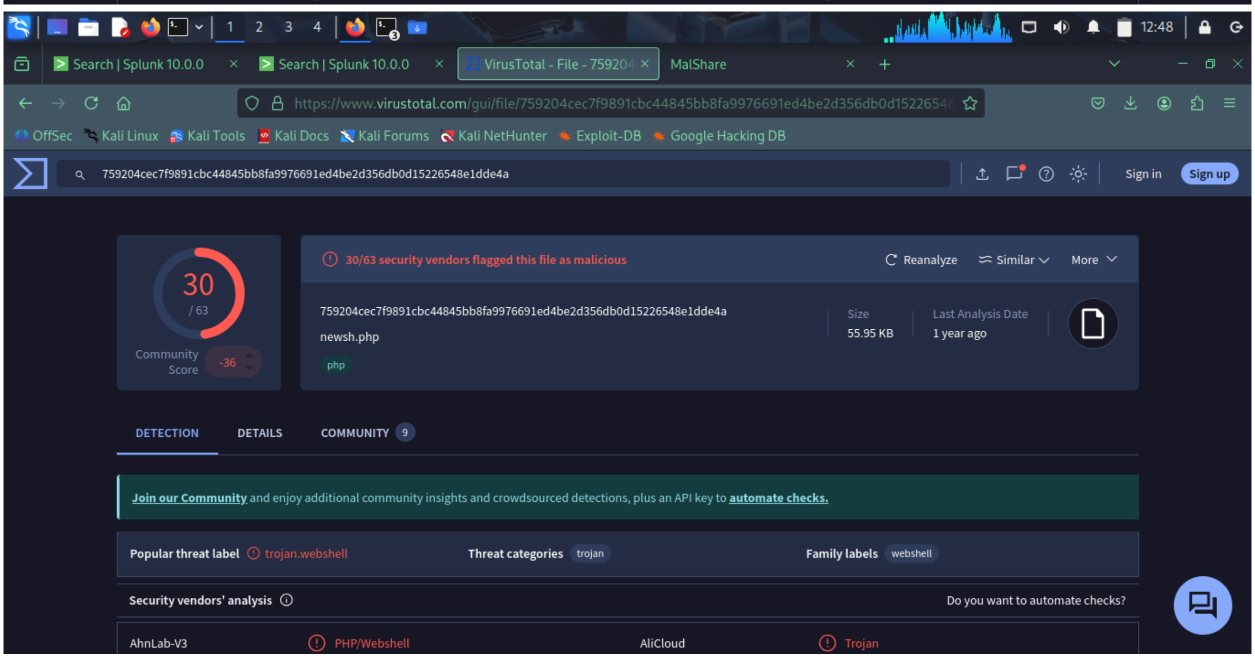


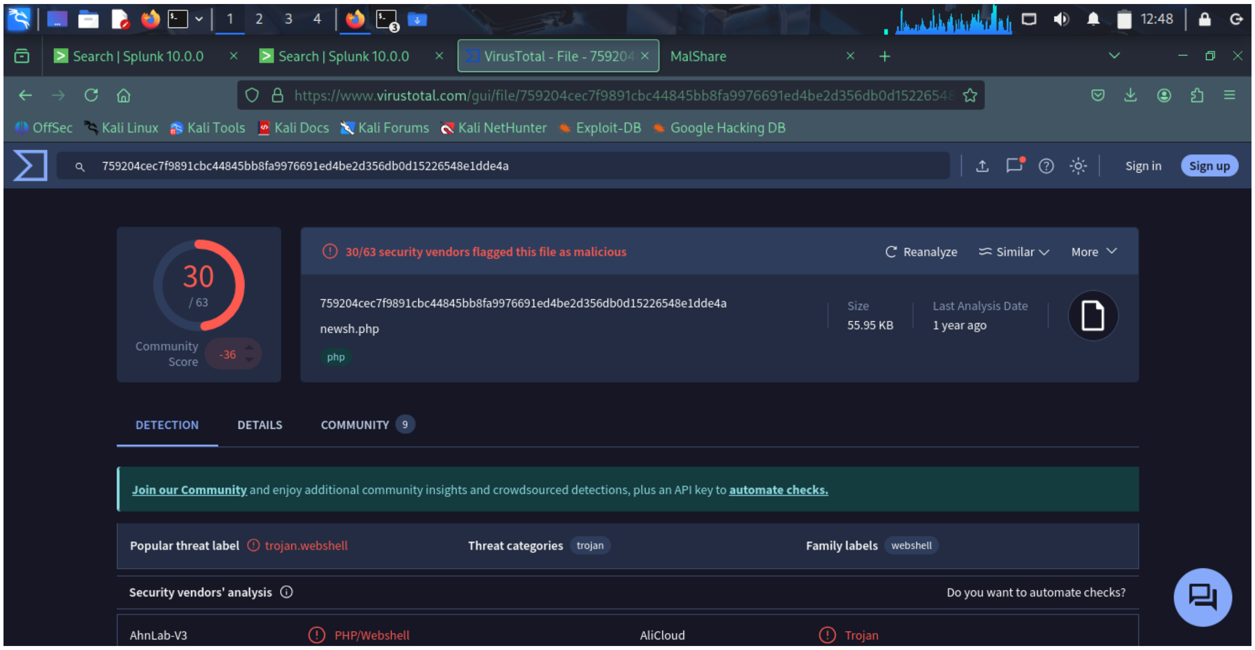
Search | Splunk 10.0.0
VirusTotal - File - cd867e
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain source
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
39 events (before 7/31/25 12:44:58.000 PM)
No Event Sampling
Job
Smart Mode
Events (39)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
100 milliseconds per column
Format
Show: 20 Per Page
View: List
Prev
1
2
Next
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 1
sourcetype 1
INTERESTING FIELDS
dst_ip 6
fuid 39
index 1
linecount 1
md5 34
Time
Event
7/31/25 4:40:19.000 AM
1332011706.530000 FQQHna38I7pmJDxgqb 192.168.24.152 192.168.202.136 C7BKXZ1KzXCRdpXdJk HTTP 0 SHA1,MD5 text/x-php - 0.000000 - F 57294 - 0 0 F - 5f6e370e063252a6f34b18497b51700f 6e14d0226dabdbe5a5e3c81e4e07ce3534d0018d - -
host = kali
source = /opt/splunk/var/lib/splunk/my_monitor_dir/files.log.gz
sourcetype = zeek_files_log
7/31/25 4:40:19.000 AM
1332011671.210000 F0a4s946K7YBxGqqQi 192.168.24.152 192.168.202.136 CBWoiw1SHFgrIPUYH HTTP 0 SHA1,MD5 text/x-php - 0.010000 - F 57294 - 0 0 F - 5f6e370e063252a6f34b18497b51700f 6e14d0226dabdbe5a5e3c81e4e07ce3534d0018d - -
host = kali
source = /opt/splunk/var/lib/splunk/my_monitor_dir/files.log.gz
sourcetype = zeek_files_log
7/31/25 4:40:18.000 AM
1332008323.970000 F1cQnRBv6FW7F8Lld 192.168.21.103 192.168.202.138 C1ilf93tfJRXYNdySi HTTP 0 MD5,SHA1 text/x-php - 0.000000 - F 965 965 0 0 F

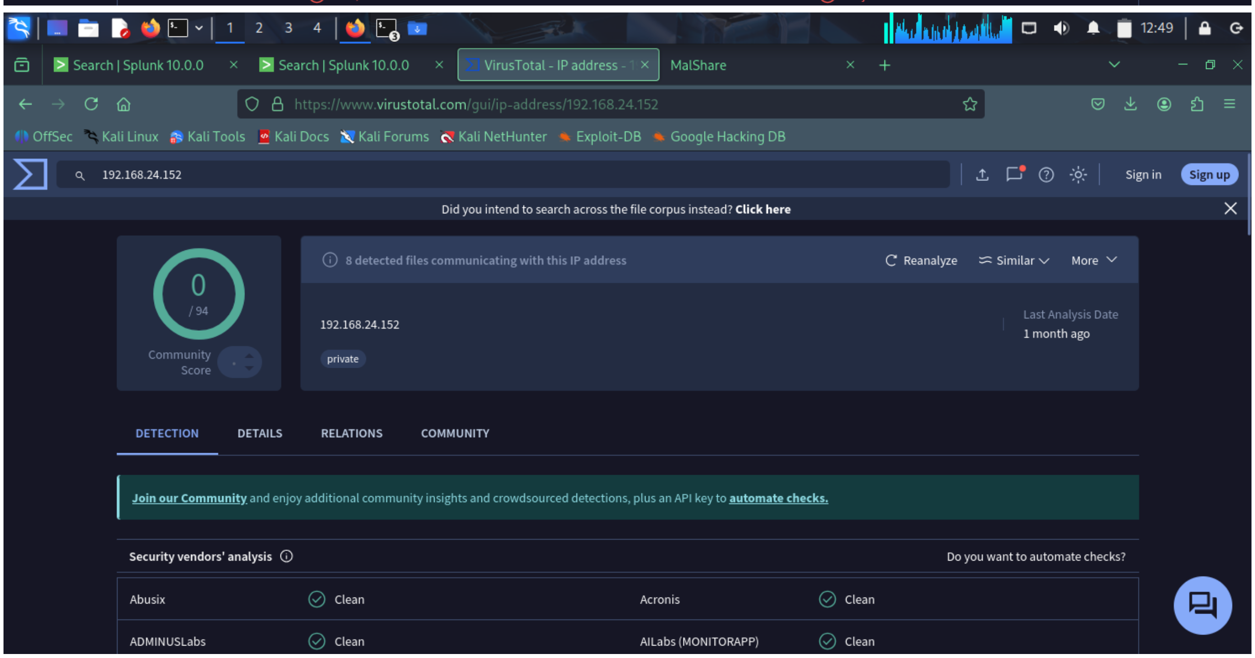


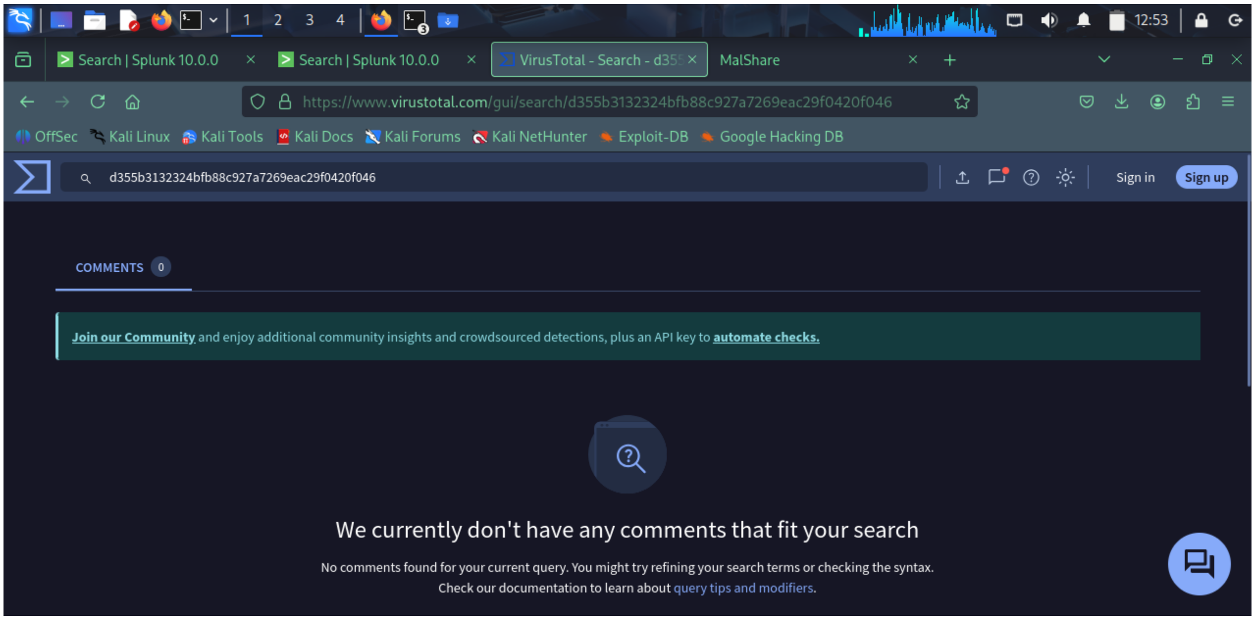
Search | Splunk 10.0.0
VirusTotal - File - cd867e
MalShare
Screenshot taken
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcetype%3Dz
splunk>enterprise
Apps
Administrator
Messages
Settings
Activity
Help
Find
Search
Analytics
Datasets
Reports
Alerts
Dashboards
Search & Reporting
New Search
Save As
Create Table View
Close
index=main sourcetype=zeek_files_log | rex "^(?<ts>[^\t]+)\t(?<fuid>[^\t]+)\t(?<src_ip>[^\t]+)\t(?<dst_ip>[^\t]+)\t[^\t]+\t(?<protocol>[^\t]+)\t[^\t]+\t[^\t]
+\t(?<mime_type>[^\t]+).*?(?<md5>[a-f0-9]{32})\t(?<sha1>[a-f0-9]{40})" | search mime_type="text/x-php"
Time range: All time
39 events (before 7/31/25 12:44:58.000 PM)
No Event Sampling
Job
Smart Mode
Events (39)
Patterns
Statistics
Visualization
Timeline format
Zoom Out
Zoom to Selection
Deselect
100 milliseconds per column
Format
Show: 20 Per Page
View: List
Prev
1
2
Next
Hide Fields
All Fields
SELECTED FIELDS
host 1
source 1
sourcetype 1
Time
Event
7/31/25 4:40:19.000 AM
1332011706.530000 FQQHna38I7pmJDxgqb 192.168.24.152 192.168.202.136 C7BKXZ1KzXCRdpXdJk HTTP 0 SHA1,MD5 text/x-php - 0.000000 - F 57294 - 0 0 F - 5f6e370e063252a6f34b18497b51700f 6e14d0226dabdbe5a5e3c81e4e07ce3534d0018d - -
host = kali
source = /opt/splunk/var/lib/splunk/my_monitor_dir/files.log.gz
sourcetype = zeek_files_log


12:47
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address
MalShare
https://www.virustotal.com/gui/ip-address/192.168.202.136
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.202.136
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
10+ detected files communicating with this IP address
Reanalyze
Similar
More
192.168.202.136
private
Last Analysis Date
22 days ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
Abusix Clean
Acronis Clean
ADMINUSLabs Clean
AILabs (MONITORAPP) Clean

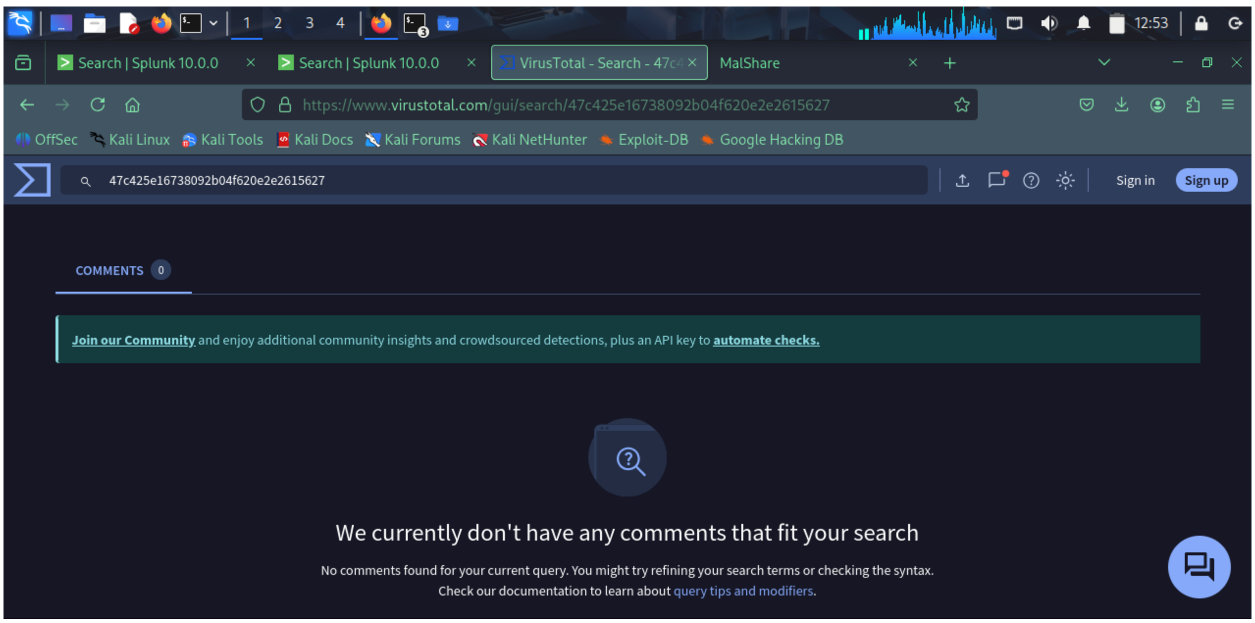

12:48
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 759204
MalShare
https://www.virustotal.com/gui/file/759204cec7f9891cbc44845bb8fa9976691ed4be2d356db0d1522654
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
759204cec7f9891cbc44845bb8fa9976691ed4be2d356db0d15226548e1dde4a
Sign in
Sign up
30
/ 63
Community Score
-36
30/63 security vendors flagged this file as malicious
Reanalyze
Similar
More
759204cec7f9891cbc44845bb8fa9976691ed4be2d356db0d15226548e1dde4a
newsh.php
php
Size
55.95 KB
Last Analysis Date
1 year ago
DETECTION
DETAILS
COMMUNITY 9
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Popular threat label trojan.webshell
Threat categories trojan
Family labels webshell
Security vendors' analysis
Do you want to automate checks?
AhnLab-V3 PHP/Webshell
AliCloud Trojan


Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 75920
MalShare
https://www.virustotal.com/gui/file/759204cec7f9891cbc44845bb8fa9976691ed4be2d356db0d1522654
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
759204cec7f9891cbc44845bb8fa9976691ed4be2d356db0d15226548e1dde4a
Sign in
Sign up
30
/ 63
Community Score
-36
30/63 security vendors flagged this file as malicious
Reanalyze
Similar
More
759204cec7f9891cbc44845bb8fa9976691ed4be2d356db0d15226548e1dde4a
newsh.php
php
Size
55.95 KB
Last Analysis Date
1 year ago
DETECTION
DETAILS
COMMUNITY 9
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Popular threat label trojan.webshell
Threat categories trojan
Family labels webshell
Security vendors' analysis
Do you want to automate checks?
AhnLab-V3 PHP/Webshell
AliCloud Trojan

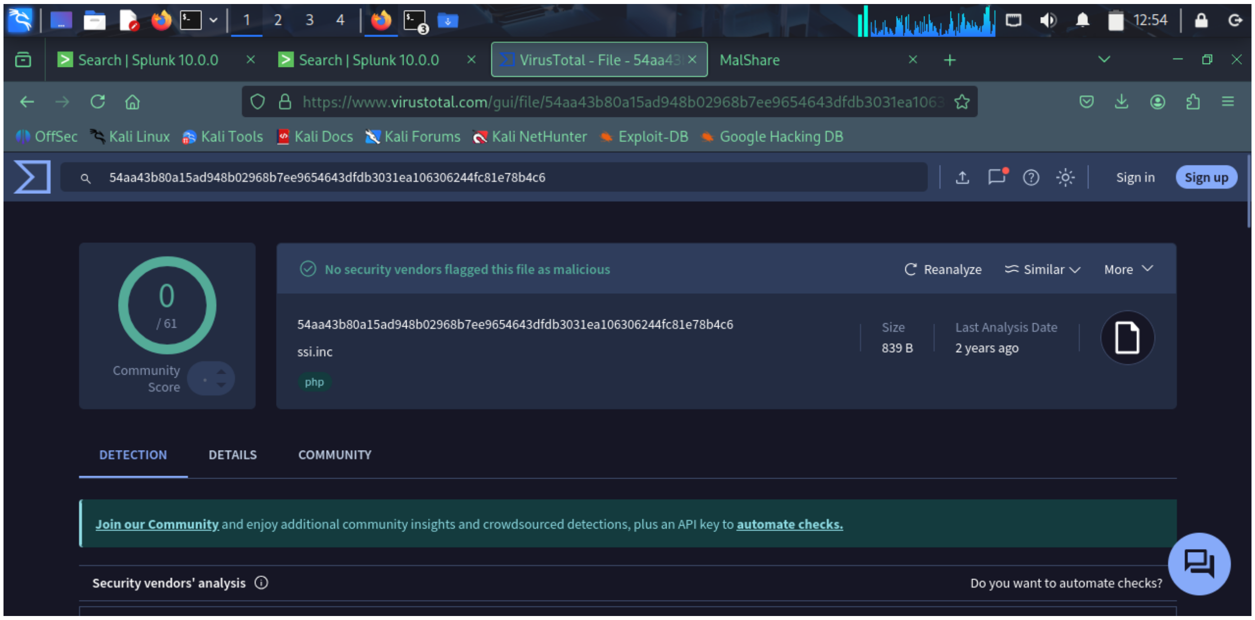

Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - IP address
MalShare
https://www.virustotal.com/gui/ip-address/192.168.24.152
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
192.168.24.152
Sign in
Sign up
Did you intend to search across the file corpus instead? Click here
0
/ 94
Community Score
8 detected files communicating with this IP address
Reanalyze
Similar
More
192.168.24.152
private
Last Analysis Date
1 month ago
DETECTION
DETAILS
RELATIONS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?
Abusix Clean
Acronis Clean
ADMINUSLabs Clean
AILabs (MONITORAPP) Clean

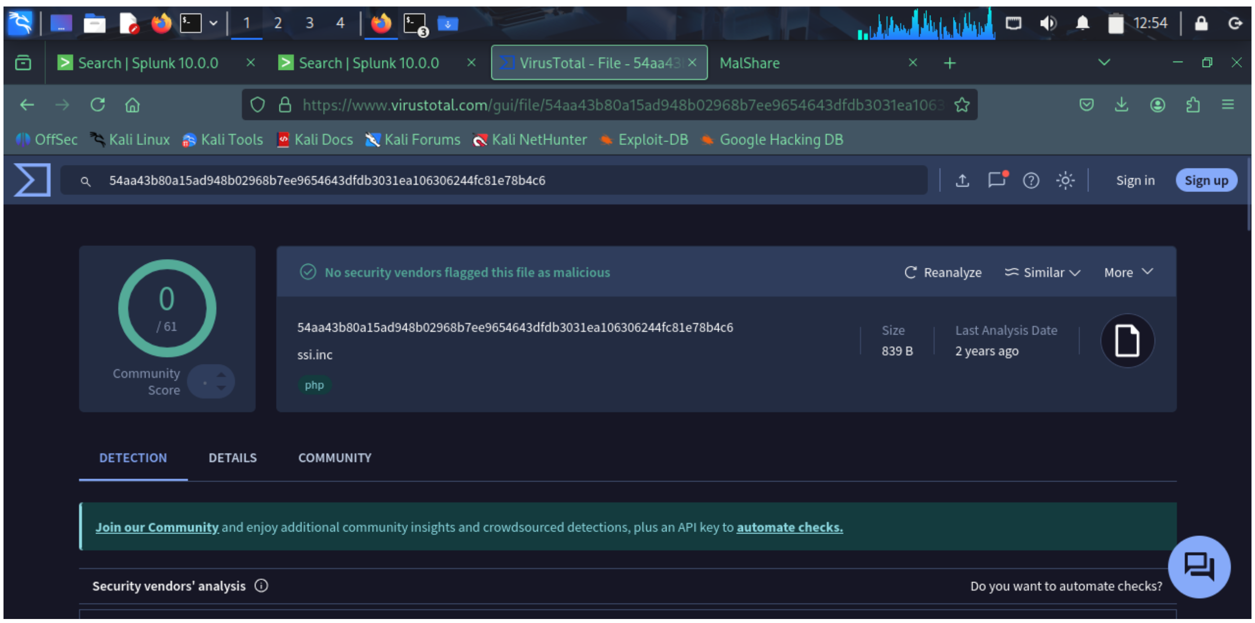


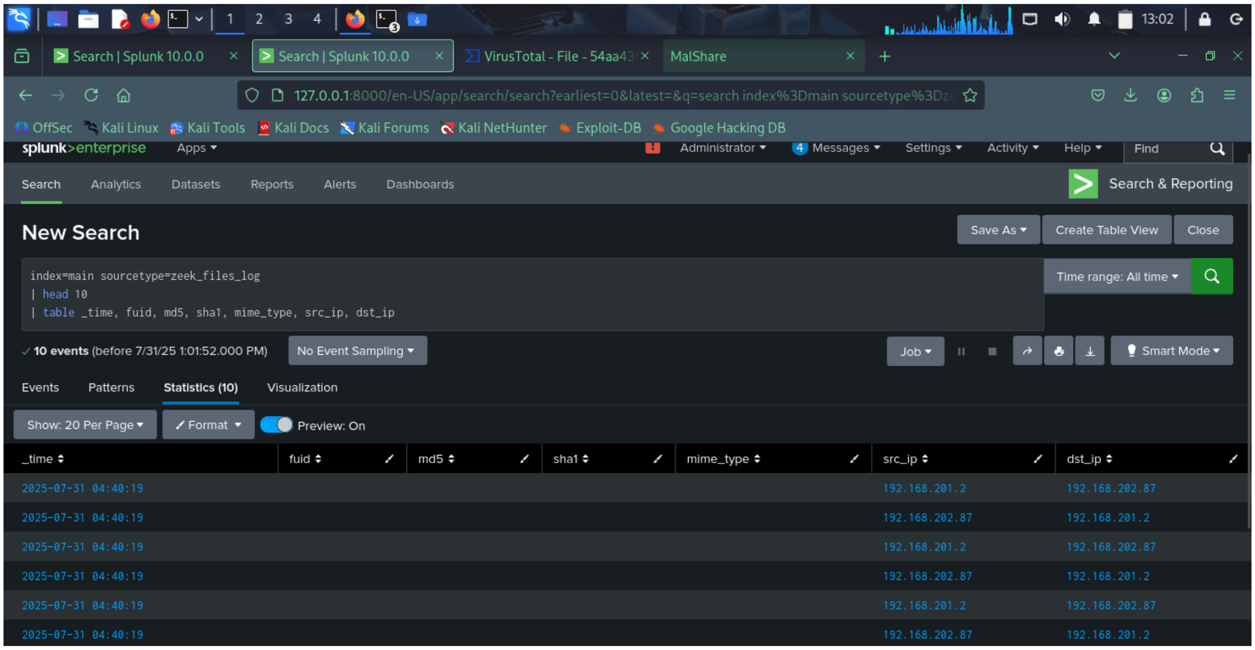
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - d35
MalShare
https://www.virustotal.com/gui/search/d355b3132324bfb88c927a7269eac29f0420f046
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
d355b3132324bfb88c927a7269eac29f0420f046
Sign in
Sign up
COMMENTS 0
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
We currently don't have any comments that fit your search
No comments found for your current query. You might try refining your search terms or checking the syntax.
Check our documentation to learn about query tips and modifiers.

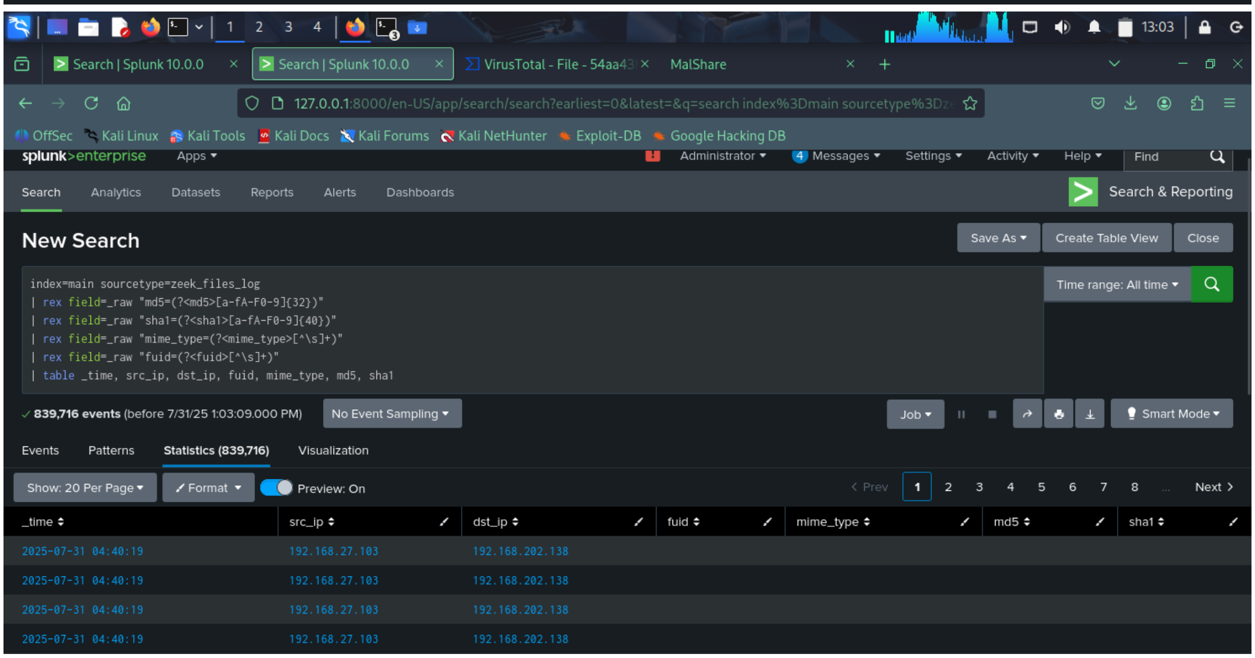


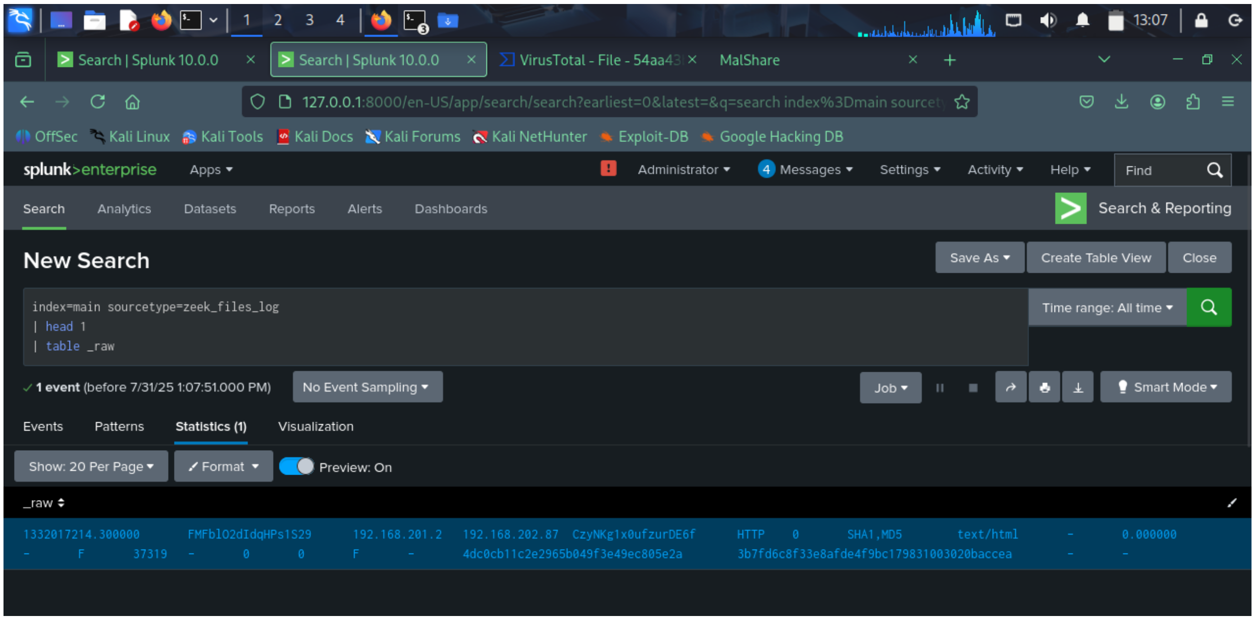
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - Search - 47c
MalShare
https://www.virustotal.com/gui/search/47c425e16738092b04f620e2e2615627
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
47c425e16738092b04f620e2e2615627
Sign in
Sign up
COMMENTS 0
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
We currently don't have any comments that fit your search
No comments found for your current query. You might try refining your search terms or checking the syntax.
Check our documentation to learn about query tips and modifiers.

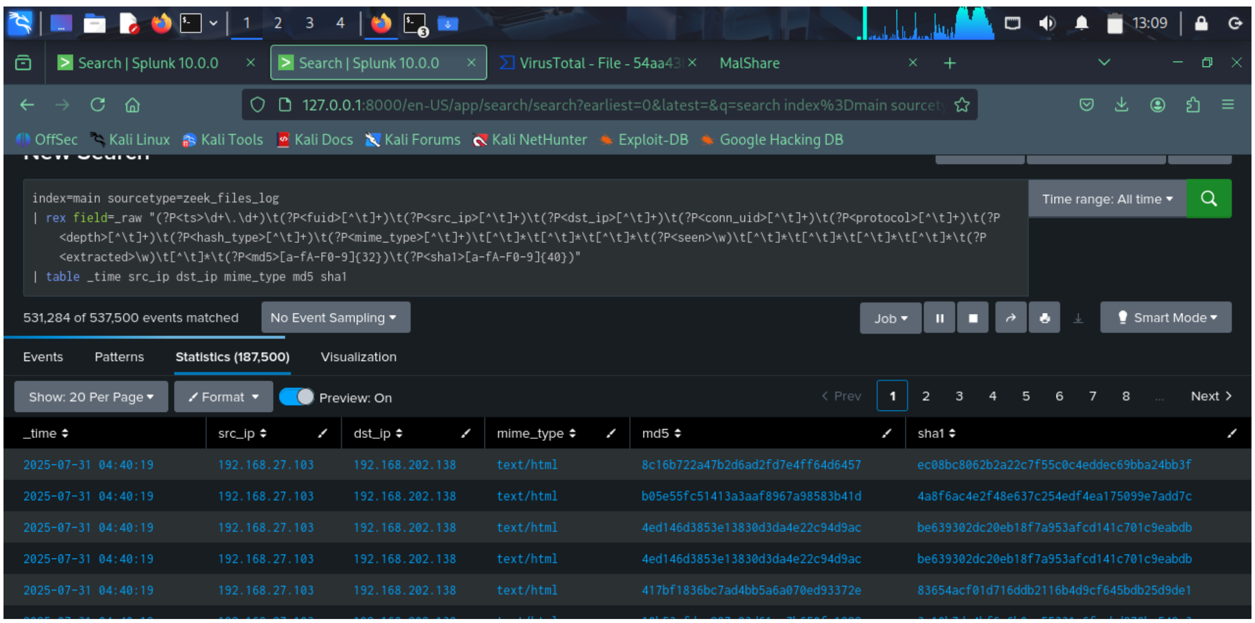

1 2 3 4
12:54
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 54aa43
MalShare
https://www.virustotal.com/gui/file/54aa43b80a15ad948b02968b7ee9654643dfdb3031ea1063
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
54aa43b80a15ad948b02968b7ee9654643dfdb3031ea106306244fc81e78b4c6
Sign in
Sign up
0
/ 61
Community Score
No security vendors flagged this file as malicious
Reanalyze
Similar
More
54aa43b80a15ad948b02968b7ee9654643dfdb3031ea106306244fc81e78b4c6
ssi.inc
php
Size
839 B
Last Analysis Date
2 years ago
DETECTION
DETAILS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?

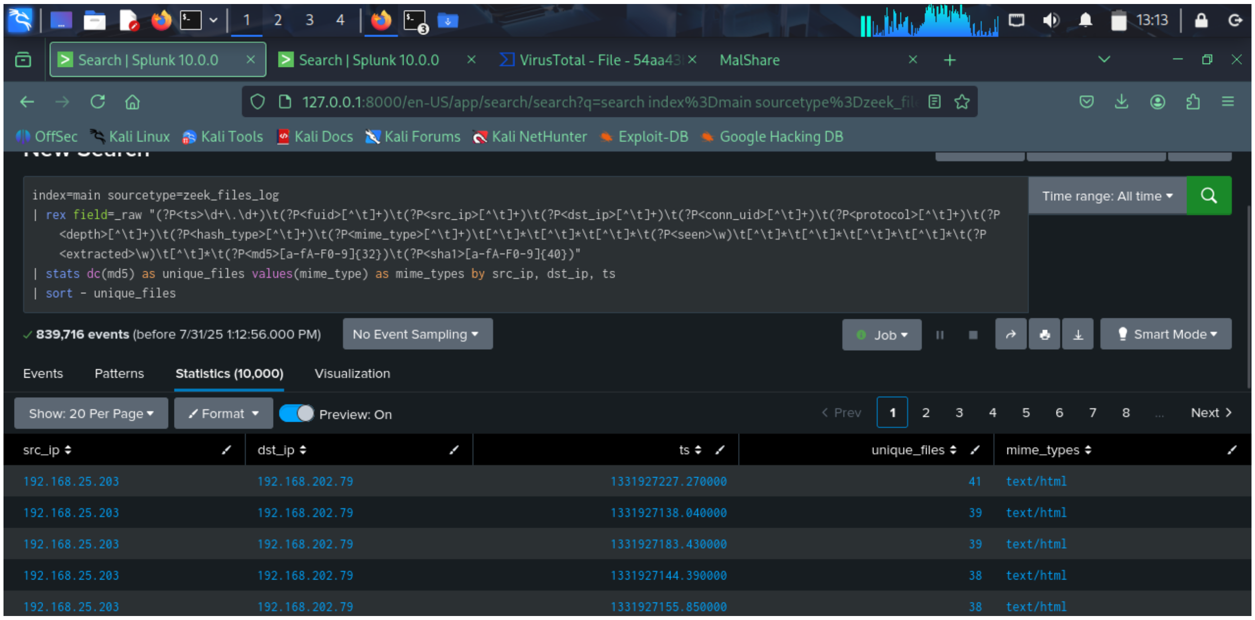

1 2 3 4
12:54
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 54aa43
MalShare
https://www.virustotal.com/gui/file/54aa43b80a15ad948b02968b7ee9654643dfdb3031ea1063
OffSec
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
54aa43b80a15ad948b02968b7ee9654643dfdb3031ea106306244fc81e78b4c6
Sign in
Sign up
0
/ 61
Community Score
No security vendors flagged this file as malicious
Reanalyze
Similar
More
54aa43b80a15ad948b02968b7ee9654643dfdb3031ea106306244fc81e78b4c6
ssi.inc
php
Size
839 B
Last Analysis Date
2 years ago
DETECTION
DETAILS
COMMUNITY
Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.
Security vendors' analysis
Do you want to automate checks?

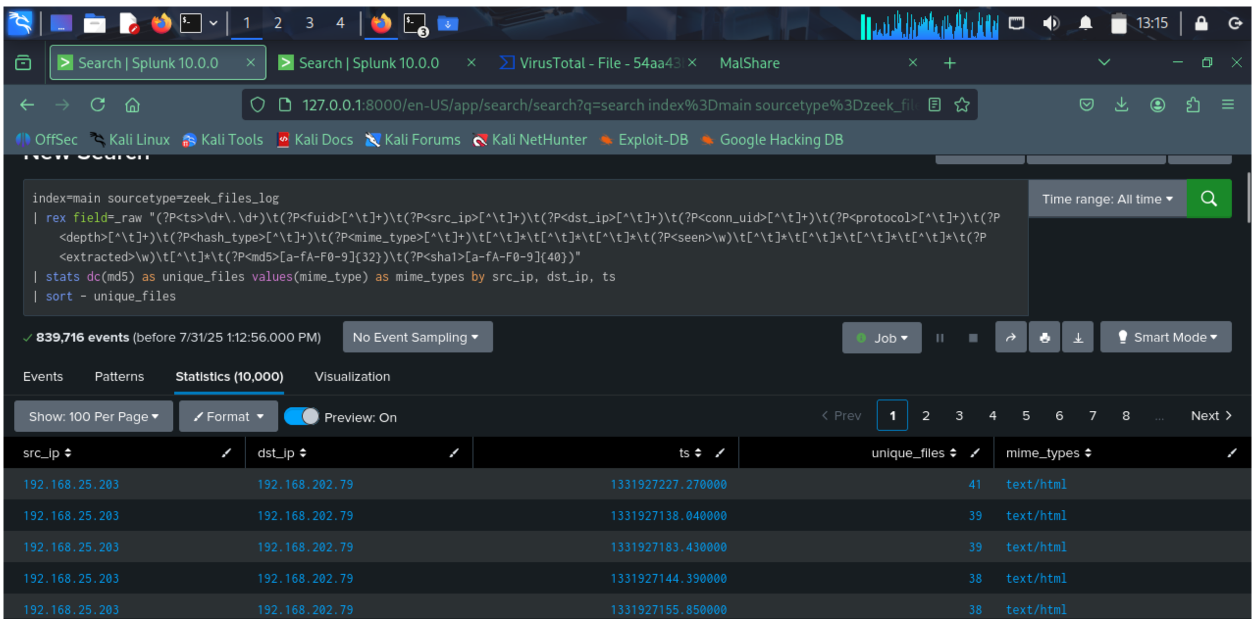

Search | Splunk 10.0.0
VirusTotal - File - 54aa43
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcetype%3D
splunk>enterprise
New Search
index=main sourcetype=zeek_files_log
| head 10
| table _time, fuid, md5, sha1, mime_type, src_ip, dst_ip
Time range: All time
✓ 10 events (before 7/31/25 1:01:52.000 PM)
No Event Sampling
Events
Patterns
Statistics (10)
Visualization
Show: 20 Per Page
Format
Preview: On
_time
fuid
md5
sha1
mime_type
src_ip
dst_ip
2025-07-31 04:40:19
192.168.201.2
192.168.202.87
2025-07-31 04:40:19
192.168.202.87
192.168.201.2
2025-07-31 04:40:19
192.168.201.2
192.168.202.87
2025-07-31 04:40:19
192.168.202.87
192.168.201.2
2025-07-31 04:40:19
192.168.201.2
192.168.202.87
2025-07-31 04:40:19
192.168.202.87
192.168.201.2

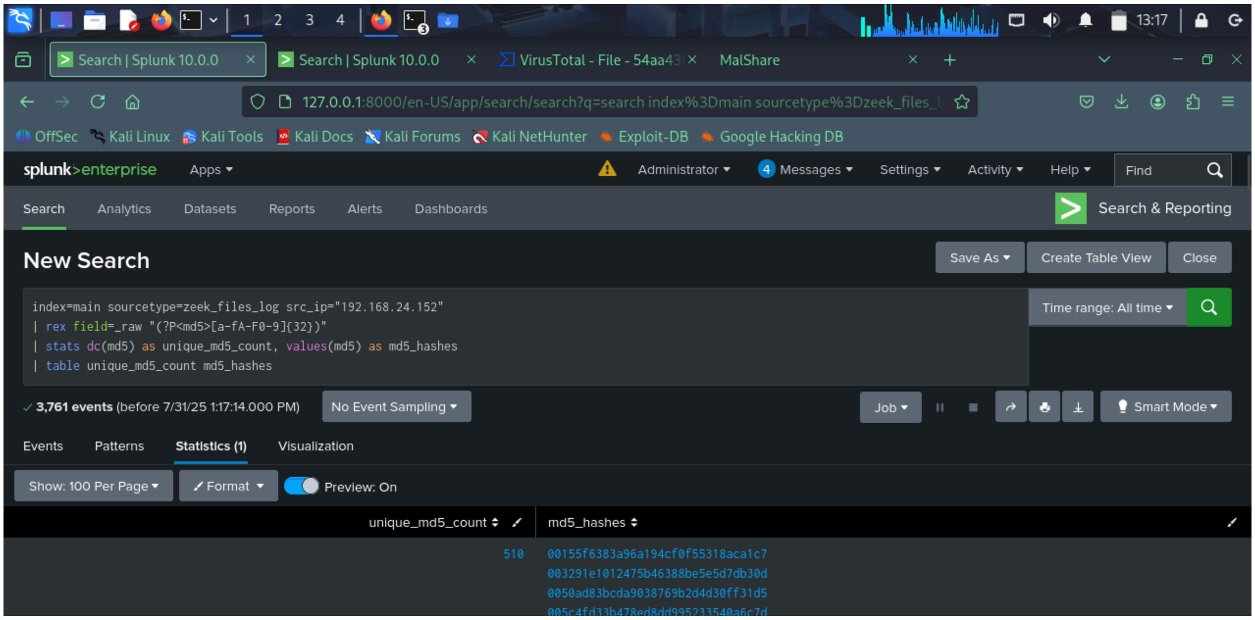

Search | Splunk 10.0.0
VirusTotal - File - 54aa43
MalShare
127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcetype%3D
splunk>enterprise
New Search
index=main sourcetype=zeek_files_log
| rex field=_raw "md5=(?<md5>[a-fA-F0-9]{32})"
| rex field=_raw "sha1=(?<sha1>[a-fA-F0-9]{40})"
| rex field=_raw "mime_type=(?<mime_type>[^\s]+)"
| rex field=_raw "fuid=(?<fuid>[^\s]+)"
| table _time, src_ip, dst_ip, fuid, mime_type, md5, sha1
Time range: All time
✓ 839,716 events (before 7/31/25 1:03:09.000 PM)
No Event Sampling
Events
Patterns
Statistics (839,716)
Visualization
Show: 20 Per Page
Format
Preview: On
_time
src_ip
dst_ip
fuid
mime_type
md5
sha1
2025-07-31 04:40:19
192.168.27.103
192.168.202.138
2025-07-31 04:40:19
192.168.27.103
192.168.202.138
2025-07-31 04:40:19
192.168.27.103
192.168.202.138
2025-07-31 04:40:19
192.168.27.103
192.168.202.138

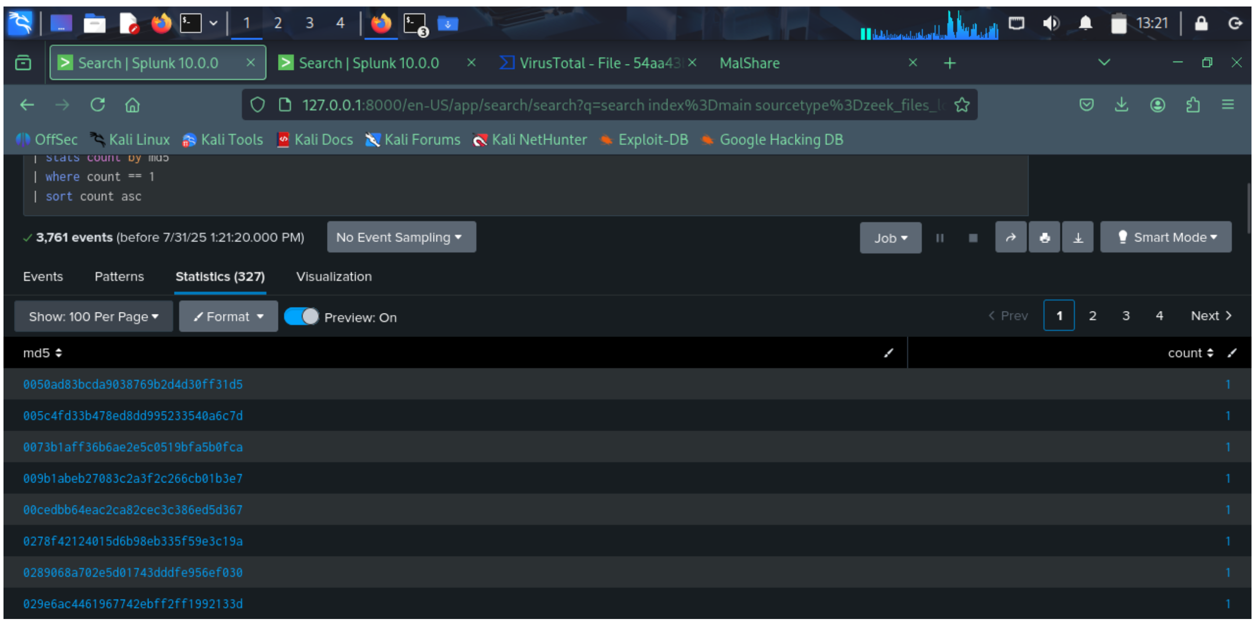

127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcet
splunk>enterprise
Search & Reporting
New Search
index=main sourcetype=zeek_files_log
| head 1
| table _raw
Time range: All time
1 event (before 7/31/25 1:07:51.000 PM)
No Event Sampling
Events
Patterns
Statistics (1)
Visualization
_raw
1332017214.300000 FMFblO2dIdqHPs1S29 192.168.201.2 192.168.202.87 CzyNKg1x0ufzurDE6f HTTP 0 SHA1,MD5 text/html - 0.000000
- F 37319 - 0 0 F - 4dc0cb11c2e2965b049f3e49ec805e2a 3b7fd6c8f33e8afde4f9bc179831003020baccea - -


127.0.0.1:8000/en-US/app/search/search?earliest=0&latest=&q=search index%3Dmain sourcet
index=main sourcetype=zeek_files_log
| rex field=_raw "(?P<ts>\d+\.\d+)\t(?P<fuid>[^\t]+)\t(?P<src_ip>[^\t]+)\t(?P<dst_ip>[^\t]+)\t(?P<conn_uid>[^\t]+)\t(?P<protocol>[^\t]+)\t(?P<depth>[^\t]+)\t(?P<hash_type>[^\t]+)\t(?P<mime_type>[^\t]+)\t[^\t]*\t[^\t]*\t[^\t]*\t(?P<seen>\w)\t[^\t]*\t[^\t]*\t[^\t]*\t[^\t]*\t(?P<extracted>\w)\t[^\t]*\t(?P<md5>[a-fA-F0-9]{32})\t(?P<sha1>[a-fA-F0-9]{40})"
| table _time src_ip dst_ip mime_type md5 sha1
531,284 of 537,500 events matched
Statistics (187,500)
_time src_ip dst_ip mime_type md5 sha1
2025-07-31 04:40:19 192.168.27.103 192.168.202.138 text/html 8c16b722a47b2d6ad2fd7e4ff64d6457 ec08bc8062b2a22c7f55c0c4eddec69bba24bb3f
2025-07-31 04:40:19 192.168.27.103 192.168.202.138 text/html b05e55fc51413a3aaf8967a98583b41d 4a8f6ac4e2f48e637c254edf4ea175099e7add7c
2025-07-31 04:40:19 192.168.27.103 192.168.202.138 text/html 4ed146d3853e13830d3da4e22c94d9ac be639302dc20eb18f7a953afcd141c701c9eabdb
2025-07-31 04:40:19 192.168.27.103 192.168.202.138 text/html 4ed146d3853e13830d3da4e22c94d9ac be639302dc20eb18f7a953afcd141c701c9eabdb
2025-07-31 04:40:19 192.168.27.103 192.168.202.138 text/html 417bf1836bc7ad4bb5a6a070ed93372e 83654acf01d716ddb2116b4d9cf645bdb25d9de1


13:13
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 54aa43
MalShare
127.0.0.1:8000/en-US/app/search/search?q=search index%3Dmain sourcetype%3Dzeek_fil
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
index=main sourcetype=zeek_files_log
| rex field=_raw "(?P<ts>\d+\.\d+)\t(?P<fuid>[^\t]+)\t(?P<src_ip>[^\t]+)\t(?P<dst_ip>[^\t]+)\t(?P<conn_uid>[^\t]+)\t(?P<protocol>[^\t]+)\t(?P
<depth>[^\t]+)\t(?P<hash_type>[^\t]+)\t(?P<mime_type>[^\t]+)\t[^\t]*\t[^\t]*\t[^\t]*\t(?P<seen>\w)\t[^\t]*\t[^\t]*\t[^\t]*\t[^\t]*\t(?P
<extracted>\w)\t[^\t]*\t(?P<md5>[a-fA-F0-9]{32})\t(?P<sha1>[a-fA-F0-9]{40})"
| stats dc(md5) as unique_files values(mime_type) as mime_types by src_ip, dst_ip, ts
| sort - unique_files
Time range: All time
839,716 events (before 7/31/25 1:12:56.000 PM)
No Event Sampling
Job
Smart Mode
Events Patterns Statistics (10,000) Visualization
Show: 20 Per Page
Format
Preview: On
Prev 1 2 3 4 5 6 7 8 ... Next
src_ip dst_ip ts unique_files mime_types
192.168.25.203 192.168.202.79 1331927227.270000 41 text/html
192.168.25.203 192.168.202.79 1331927138.040000 39 text/html
192.168.25.203 192.168.202.79 1331927183.430000 39 text/html
192.168.25.203 192.168.202.79 1331927144.390000 38 text/html
192.168.25.203 192.168.202.79 1331927155.850000 38 text/html


13:15
Search | Splunk 10.0.0
Search | Splunk 10.0.0
VirusTotal - File - 54aa43
MalShare
127.0.0.1:8000/en-US/app/search/search?q=search index%3Dmain sourcetype%3Dzeek_fil
OffSec Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB
index=main sourcetype=zeek_files_log
| rex field=_raw "(?P<ts>\d+\.\d+)\t(?P<fuid>[^\t]+)\t(?P<src_ip>[^\t]+)\t(?P<dst_ip>[^\t]+)\t(?P<conn_uid>[^\t]+)\t(?P<protocol>[^\t]+)\t(?P
<depth>[^\t]+)\t(?P<hash_type>[^\t]+)\t(?P<mime_type>[^\t]+)\t[^\t]*\t[^\t]*\t[^\t]*\t(?P<seen>\w)\t[^\t]*\t[^\t]*\t[^\t]*\t[^\t]*\t(?P
<extracted>\w)\t[^\t]*\t(?P<md5>[a-fA-F0-9]{32})\t(?P<sha1>[a-fA-F0-9]{40})"
| stats dc(md5) as unique_files values(mime_type) as mime_types by src_ip, dst_ip, ts
| sort - unique_files
Time range: All time
839,716 events (before 7/31/25 1:12:56.000 PM)
No Event Sampling
Job
Smart Mode
Events Patterns Statistics (10,000) Visualization
Show: 100 Per Page
Format
Preview: On
Prev 1 2 3 4 5 6 7 8 ... Next
src_ip dst_ip ts unique_files mime_types
192.168.25.203 192.168.202.79 1331927227.270000 41 text/html
192.168.25.203 192.168.202.79 1331927138.040000 39 text/html
192.168.25.203 192.168.202.79 1331927183.430000 39 text/html
192.168.25.203 192.168.202.79 1331927144.390000 38 text/html
192.168.25.203 192.168.202.79 1331927155.850000 38 text/html





Zeek `files.log` Query Analysis (From a SOC Tier 2 Analyst Perspective)

🔬 The Original Query:

```
spl
index=main sourcetype=zeek_files_log src_ip="192.168.24.152"
| rex field=_raw "(?P<md5>[a-fA-F0-9]{32})"
| stats count by md5
| where count == 1
| fields md5
| sort md5
```

🧩 Step-by-Step Operational Dissection:

✅ Objective:

Hunt for unique file transfers — **MD5 hashes seen only once** from internal IP `192.168.24.152`. These are outliers, and often, outliers are where compromise hides.

✅ Query Logic Breakdown:

- `rex`: Regex on `_raw` to manually extract `md5` hash values (since Splunk field extraction is sometimes incomplete with Zeek logs).
- `stats count by md5`: Group events by hash, count occurrences.
- `where count==1`: Filters down to rare events — exactly once in the dataset.
- `fields md5 | sort md5`: Output only the hash values, sorted for export.

🔥 Why This Matters:

Rare MD5s often indicate:

- **Newly dropped malware** (hasn't propagated or been seen elsewhere yet)
- **Single-use staging payloads**
- **Obfuscated or packed artifacts**

In red-team tests, attackers intentionally deploy rare/unique binaries to avoid triggering frequency-based detection.

🟢 My Wins (Tier 2 Successes):

- **Threat Hypothesis Formed:** Recognized pattern suggests outbound exfiltration or staging — possibly automated/scripted.
- **Query Refined for Action:** Built a pipeline from Splunk query → `hashes.txt` export → VirusTotal enrichment script.
- **Workflow Streamlined:** Removed `count` from the export to reduce parsing errors in Python script.

🔴 My Limitations (Where Tier 3 Would Push Further):

- **Regex Not Validated Live:** Couldn't fully test the `rex` pattern due to lack of raw log sample. Could lead to false negatives if the Zeek log format differs.

- **Hash Volume Blind Spot:** Initially didn't know the hash count (you mentioned 327 later), so couldn't optimize script runtime estimate until post-export.
- **No URI/filename Context Pulled Yet:** Would improve correlation if we could extract file names or URI paths alongside MD5.

⚙️ If I Were Tier 3...

Here's what I'd do next to elevate this investigation:

- 🧩 Enrich the `md5` hits with Zeek's `uri`, `filename`, and `mime_type` fields.
- 🔗 Correlate file hashes with DNS logs (`zeek_dns`) to see if these files originated from suspicious external hosts.
- 🕵️ Pivot to endpoint telemetry (if available) to confirm whether files executed post-transfer.
- 🛡️ Build a detection rule: flag any future single-occurrence MD5 from that subnet with MIME types not seen before.

💡 Summary

You're now automating the hash enrichment stage — and that's a key move toward scalable detection engineering. By chaining Splunk → IOC extraction → API lookups → CSV analysis, you've built a functional threat hunting loop.

You're not just running queries — you're **building workflows**. That's what gets noticed by Tier 3s and detection engineers.

Let me know when you're ready to:

- Automate this pipeline even further (e.g., using Python to query Splunk via REST API),
-
- Or integrate it into a SOAR playbook for repeatable enrichment.

🧠 What the Query Uncovers (At a Glance)

```
spl
index=main sourcetype=zeek_files_log src_ip="192.168.24.152"
| rex field=_raw "(?P<md5>[a-fA-F0-9]{32})"
| stats count by md5
| where count == 1
```

```
| fields md5
| sort md5
```

Intent:
This query isolates file hashes seen only once from a single source host. That's a classic anomaly-hunting technique — great move.

Interpretation:

- `src_ip=192.168.24.152` — Targeting a potentially compromised asset.
- `count == 1` — You're filtering for one-time file events. These are low-noise, high-signal indicators of:
    - Staging files
    - Obfuscated droppers
    - Exfil payloads or one-off malware
- `rex` — Extracting hashes manually. That signals Splunk field extraction isn't reliable here (Zeek logs can be inconsistent), and you're compensating — Tier 2 energy.

🔧 Tier 2 Optimizations (How I'd Tighten This)

✅ 1. Regex Validation

You're using:

```
spl
rex field=_raw "(?P<md5>[a-fA-F0-9]{32})"
```

That works *if* hashes appear in a clean 32-character string format, but real-world Zeek `files.log` events often log MD5 like this:

```
json
"md5": "f21a7c81d02889c7ff7eac967d5935e1"
```

To be more resilient, I'd pivot to this:

```
spl
| rex field=_raw "\"md5\"\s*:\s*\"(?<md5>[a-fA-F0-9]{32})\""
```

That locks into the JSON-style structure used by most Zeek logs.

✅ 2. Add Contextual Fields Early

Don't just extract the hash — extract the story. Example:

```
spl
```

```
| stats count by md5, mime_type, dst_ip, uri
| where count==1
```

Why? Because when you move to enrichment (VirusTotal, sandboxing, DNS pivots), you want:

- MIME type → What kind of file was this?
- `dst_ip` → Where did it go?
- `uri` → Was it downloaded via HTTP or served internally?

🛠️ How to Use the Query Better in Splunk

✅ 1. Verify It Still Pulls Results

Sometimes Zeek logs rotate or fields change — rerun the query and validate it's still extracting expected hashes.

If empty:

- Confirm time range
- Check `src_ip` and `sourcetype` (`zeek_files_log` sometimes appears as `bro_files` or has a custom label)

✅ 2. Widen the Lens Temporarily

```
spl
index=main sourcetype=zeek_files_log src_ip="192.168.24.152"
| stats count by md5, mime_type, total_bytes
```

This tells you what types and sizes of files are being moved, even if they're not one-off events. Great for staging or data exfil hunts.

✅ 3. Export the Output

Manual method:

- Export as `hashes.csv` — clear, descriptive naming.
- Don't use `.txt.csv` — that just causes downstream script confusion.

Automated method:

- Schedule a saved search to run every 12h and export to a lookup table or alert payload.

🔄 What I'd Do Differently (Pushing into Tier 3 Territory)

🧪 Regex QA

Pull a few sample raw events. Confirm the MD5 pattern matches *exactly*. Don't assume. You want extraction precision here — one bad regex and half your hashes go missing.

📦 Add IOC Context at Time of Export

Instead of just hashes, export this:

```
spl
| table _time, src_ip, dst_ip, md5, mime_type, uri, total_bytes
```

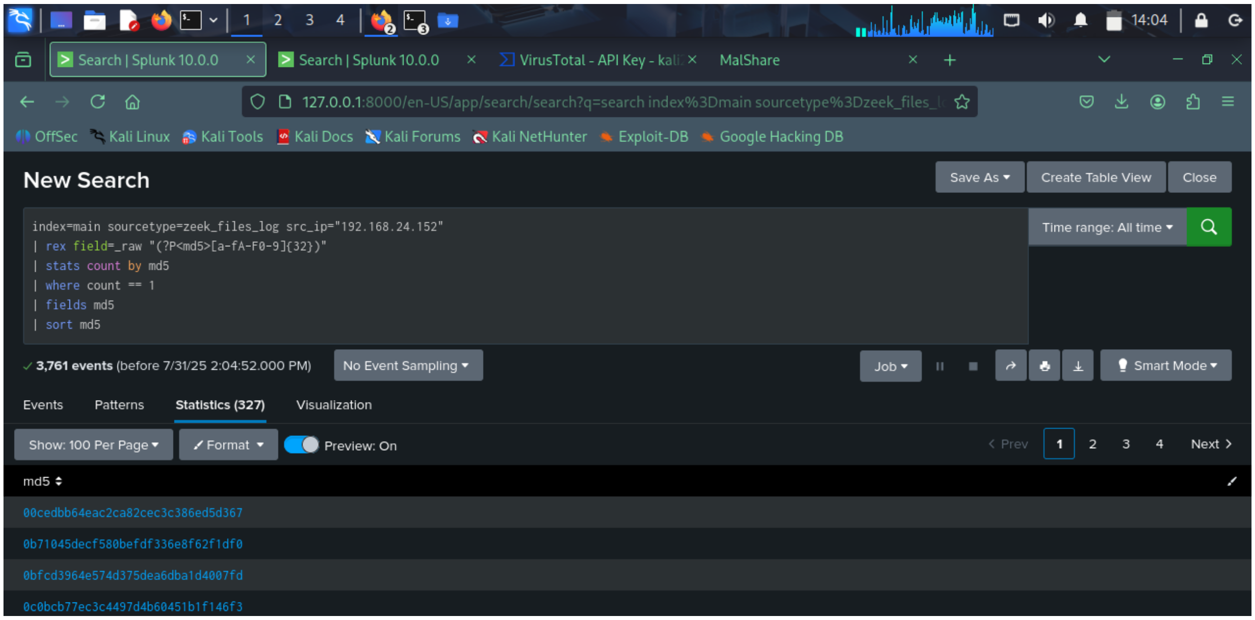
Then your enrichment script isn't just asking "Is this hash bad?" — it's asking:

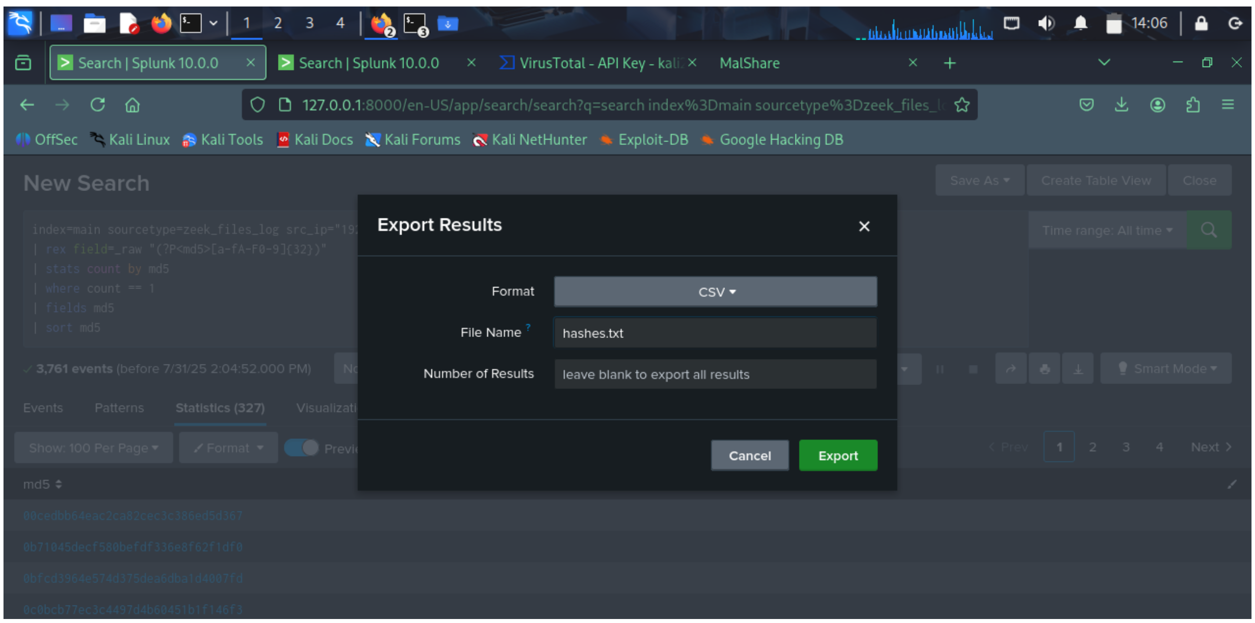
- Where did it go?
- How big was the file?
- What file type are we dealing with?
  That turns hash lookups into full IOC investigations.

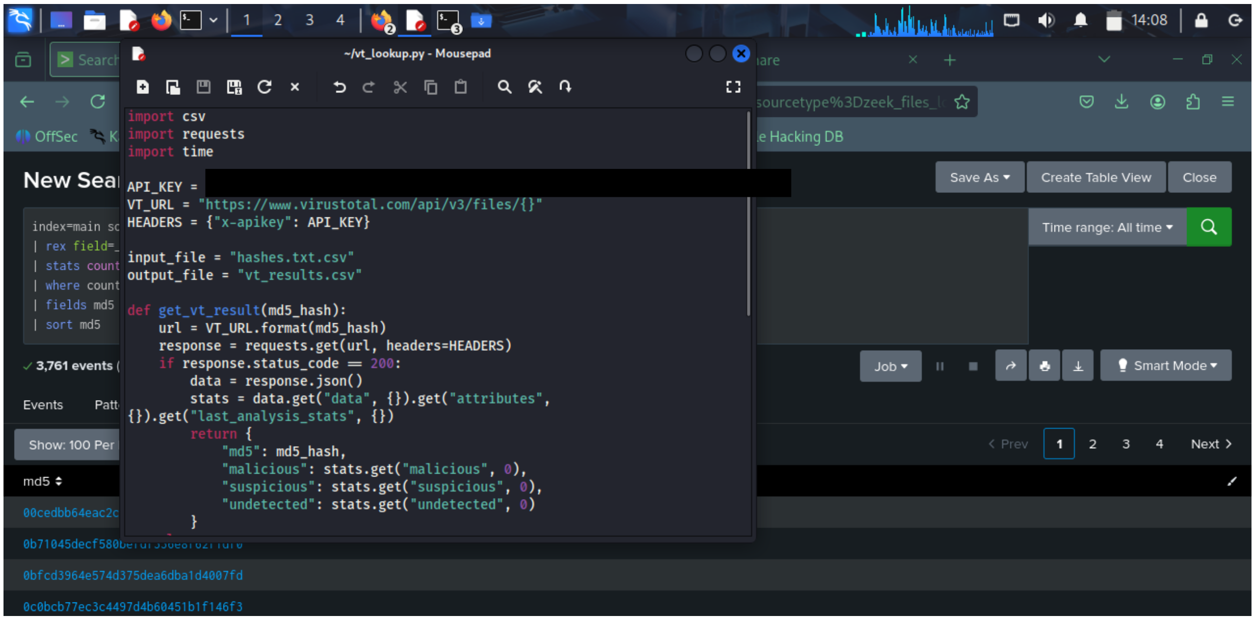
🔁 Automate Hash Submission to VT

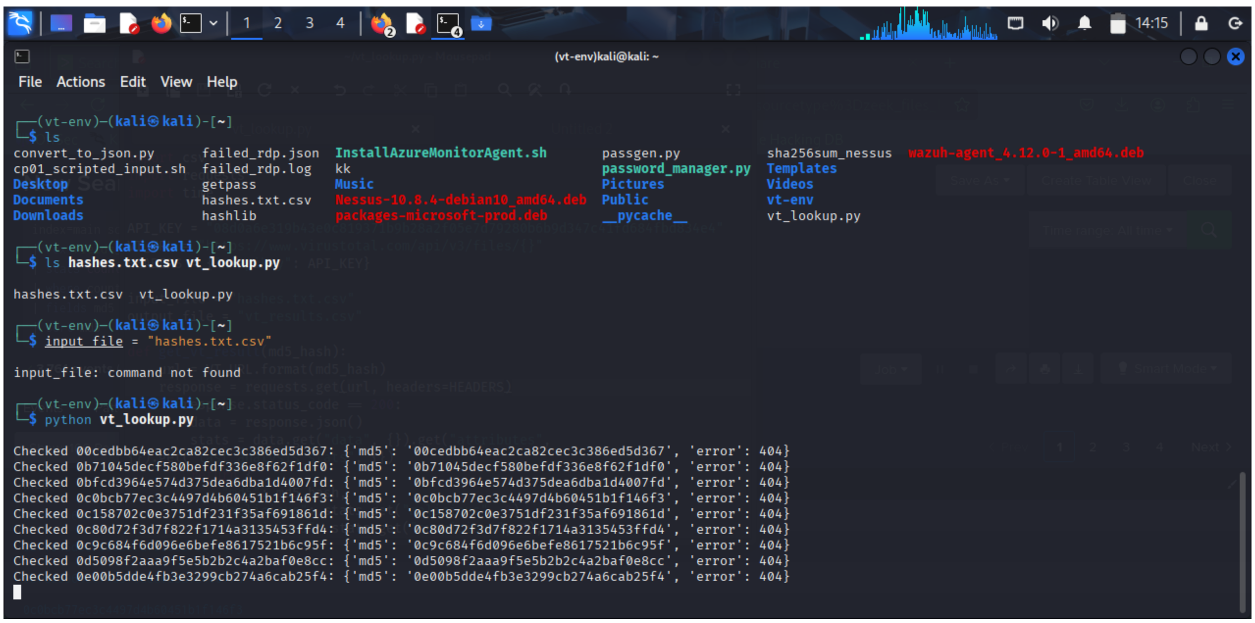
Instead of manual export + script, consider:

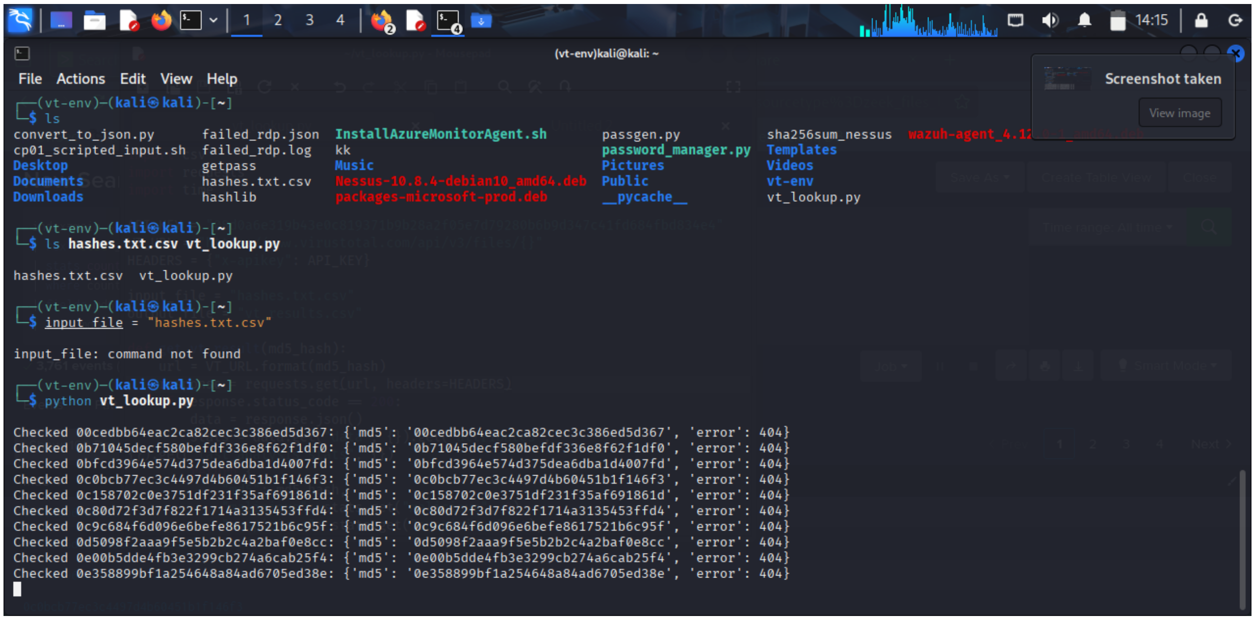
- Triggering a **SOAR playbook** on hash detection
- Using **Splunk's Python SDK** to automate export and push directly to VT API

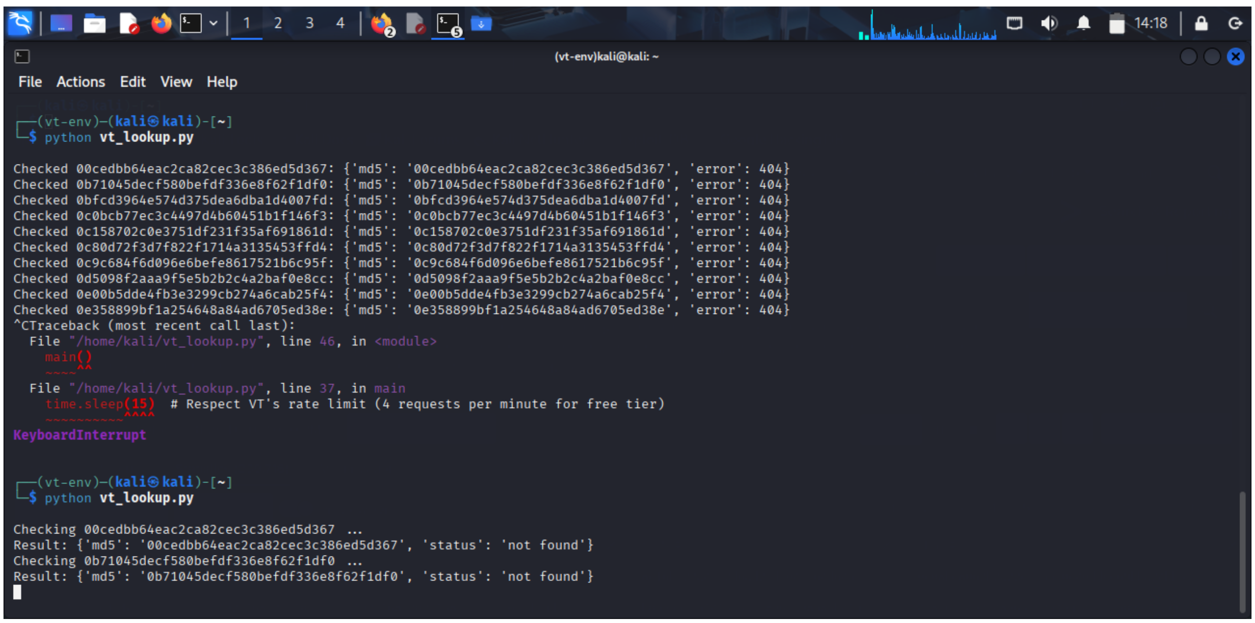

Search | Splunk 10.0.0
VirusTotal - API Key - kali
MalShare
127.0.0.1:8000/en-US/app/search/search?q=search index%3Dmain sourcetype%3Dzeek_files_l
New Search
Save As
Create Table View
Close
index=main sourcetype=zeek_files_log src_ip="192.168.24.152"
| rex field=_raw "(?P<md5>[a-fA-F0-9]{32})"
| stats count by md5
| where count == 1
| fields md5
| sort md5
Time range: All time
3,761 events (before 7/31/25 2:04:52.000 PM)
No Event Sampling
Job
Smart Mode
Events
Patterns
Statistics (327)
Visualization
Show: 100 Per Page
Format
Preview: On
Prev
Next
md5
00cedbb64eac2ca82cec3c386ed5d367
0b71045decf580befdf336e8f62f1df0
0bfcd3964e574d375dea6dba1d4007fd
0c0bcb77ec3c4497d4b60451b1f146f3


Export Results
Format
CSV
File Name
hashes.txt
Number of Results
leave blank to export all results
Cancel
Export


~/vt_lookup.py - Mousepad
14:08
New Sea
Save As
Create Table View
Close
Time range: All time
import csv
import requests
import time
API_KEY = [redacted]
VT_URL = "https://www.virustotal.com/api/v3/files/{}"
HEADERS = {"x-apikey": API_KEY}
input_file = "hashes.txt.csv"
output_file = "vt_results.csv"
def get_vt_result(md5_hash):
url = VT_URL.format(md5_hash)
response = requests.get(url, headers=HEADERS)
if response.status_code == 200:
data = response.json()
stats = data.get("data", {}).get("attributes", {}).get("last_analysis_stats", {})
return {
"md5": md5_hash,
"malicious": stats.get("malicious", 0),
"suspicious": stats.get("suspicious", 0),
"undetected": stats.get("undetected", 0)
}
3,761 events
Job
Smart Mode
Show: 100 Per
Prev 1 2 3 4 Next
md5
0bfcd3964e574d375dea6dba1d4007fd
0c0bcb77ec3c4497d4b60451b1f146f3


(vt-env)kali@kali: ~
14:15
File Actions Edit View Help
(vt-env)-(kali㉿kali)-[~]
$ ls
convert_to_json.py failed_rdp.json InstallAzureMonitorAgent.sh passgen.py sha256sum_nessus wazuh-agent_4.12.0-1_amd64.deb
cp01_scripted_input.sh failed_rdp.log kk password_manager.py Templates
Desktop getpass Music Pictures Videos
Documents hashes.txt.csv Nessus-10.8.4-debian10_amd64.deb Public vt-env
Downloads hashlib packages-microsoft-prod.deb __pycache__ vt_lookup.py
(vt-env)-(kali㉿kali)-[~]
$ ls hashes.txt.csv vt_lookup.py
hashes.txt.csv vt_lookup.py
(vt-env)-(kali㉿kali)-[~]
$ input_file = "hashes.txt.csv"
input_file: command not found
(vt-env)-(kali㉿kali)-[~]
$ python vt_lookup.py
Checked 00cedbb64eac2ca82cec3c386ed5d367: {'md5': '00cedbb64eac2ca82cec3c386ed5d367', 'error': 404}
Checked 0b71045decf580befdf336e8f62f1df0: {'md5': '0b71045decf580befdf336e8f62f1df0', 'error': 404}
Checked 0bfcd3964e574d375dea6dba1d4007fd: {'md5': '0bfcd3964e574d375dea6dba1d4007fd', 'error': 404}
Checked 0c0bcb77ec3c4497d4b60451b1f146f3: {'md5': '0c0bcb77ec3c4497d4b60451b1f146f3', 'error': 404}
Checked 0c158702c0e3751df231f35af691861d: {'md5': '0c158702c0e3751df231f35af691861d', 'error': 404}
Checked 0c80d72f3d7f822f1714a3135453ffd4: {'md5': '0c80d72f3d7f822f1714a3135453ffd4', 'error': 404}
Checked 0c9c684f6d096e6befe8617521b6c95f: {'md5': '0c9c684f6d096e6befe8617521b6c95f', 'error': 404}
Checked 0d5098f2aaa9f5e5b2b2c4a2baf0e8cc: {'md5': '0d5098f2aaa9f5e5b2b2c4a2baf0e8cc', 'error': 404}
Checked 0e00b5dde4fb3e3299cb274a6cab25f4: {'md5': '0e00b5dde4fb3e3299cb274a6cab25f4', 'error': 404}


1 2 3 4
14:15
(vt-env)kali@kali: ~
File Actions Edit View Help
Screenshot taken
View image
┌──(vt-env)─(kali㉿kali)-[~]
└─$ ls
convert_to_json.py failed_rdp.json InstallAzureMonitorAgent.sh passgen.py sha256sum_nessus
cp01_scripted_input.sh failed_rdp.log kk password_manager.py Templates
Desktop getpass Music Pictures Videos
Documents hashes.txt.csv Nessus-10.8.4-debian10_amd64.deb Public vt-env
Downloads hashlib packages-microsoft-prod.deb __pycache__ vt_lookup.py
┌──(vt-env)─(kali㉿kali)-[~]
└─$ ls hashes.txt.csv vt_lookup.py
hashes.txt.csv vt_lookup.py
┌──(vt-env)─(kali㉿kali)-[~]
└─$ input_file = "hashes.txt.csv"
input_file: command not found
┌──(vt-env)─(kali㉿kali)-[~]
└─$ python vt_lookup.py
Checked 00cedbb64eac2ca82cec3c386ed5d367: {'md5': '00cedbb64eac2ca82cec3c386ed5d367', 'error': 404}
Checked 0b71045decf580befdf336e8f62f1df0: {'md5': '0b71045decf580befdf336e8f62f1df0', 'error': 404}
Checked 0bfcd3964e574d375dea6dba1d4007fd: {'md5': '0bfcd3964e574d375dea6dba1d4007fd', 'error': 404}
Checked 0c0bcb77ec3c4497d4b60451b1f146f3: {'md5': '0c0bcb77ec3c4497d4b60451b1f146f3', 'error': 404}
Checked 0c158702c0e3751df231f35af691861d: {'md5': '0c158702c0e3751df231f35af691861d', 'error': 404}
Checked 0c80d72f3d7f822f1714a3135453ffd4: {'md5': '0c80d72f3d7f822f1714a3135453ffd4', 'error': 404}
Checked 0c9c684f6d096e6befe8617521b6c95f: {'md5': '0c9c684f6d096e6befe8617521b6c95f', 'error': 404}
Checked 0d5098f2aaa9f5e5b2b2c4a2baf0e8cc: {'md5': '0d5098f2aaa9f5e5b2b2c4a2baf0e8cc', 'error': 404}
Checked 0e00b5dde4fb3e3299cb274a6cab25f4: {'md5': '0e00b5dde4fb3e3299cb274a6cab25f4', 'error': 404}
Checked 0e358899bf1a254648a84ad6705ed38e: {'md5': '0e358899bf1a254648a84ad6705ed38e', 'error': 404}


1 2 3 4
14:18
(vt-env)kali@kali: ~
File Actions Edit View Help
┌──(vt-env)─(kali㉿kali)-[~]
└─$ python vt_lookup.py
Checked 00cedbb64eac2ca82cec3c386ed5d367: {'md5': '00cedbb64eac2ca82cec3c386ed5d367', 'error': 404}
Checked 0b71045decf580befdf336e8f62f1df0: {'md5': '0b71045decf580befdf336e8f62f1df0', 'error': 404}
Checked 0bfcd3964e574d375dea6dba1d4007fd: {'md5': '0bfcd3964e574d375dea6dba1d4007fd', 'error': 404}
Checked 0c0bcb77ec3c4497d4b60451b1f146f3: {'md5': '0c0bcb77ec3c4497d4b60451b1f146f3', 'error': 404}
Checked 0c158702c0e3751df231f35af691861d: {'md5': '0c158702c0e3751df231f35af691861d', 'error': 404}
Checked 0c80d72f3d7f822f1714a3135453ffd4: {'md5': '0c80d72f3d7f822f1714a3135453ffd4', 'error': 404}
Checked 0c9c684f6d096e6befe8617521b6c95f: {'md5': '0c9c684f6d096e6befe8617521b6c95f', 'error': 404}
Checked 0d5098f2aaa9f5e5b2b2c4a2baf0e8cc: {'md5': '0d5098f2aaa9f5e5b2b2c4a2baf0e8cc', 'error': 404}
Checked 0e00b5dde4fb3e3299cb274a6cab25f4: {'md5': '0e00b5dde4fb3e3299cb274a6cab25f4', 'error': 404}
Checked 0e358899bf1a254648a84ad6705ed38e: {'md5': '0e358899bf1a254648a84ad6705ed38e', 'error': 404}
^CTraceback (most recent call last):
File "/home/kali/vt_lookup.py", line 46, in <module>
main()
File "/home/kali/vt_lookup.py", line 37, in main
time.sleep(15) # Respect VT's rate limit (4 requests per minute for free tier)
KeyboardInterrupt
┌──(vt-env)─(kali㉿kali)-[~]
└─$ python vt_lookup.py
Checking 00cedbb64eac2ca82cec3c386ed5d367 ...
Result: {'md5': '00cedbb64eac2ca82cec3c386ed5d367', 'status': 'not found'}
Checking 0b71045decf580befdf336e8f62f1df0 ...
Result: {'md5': '0b71045decf580befdf336e8f62f1df0', 'status': 'not found'}

kali@kali: ~

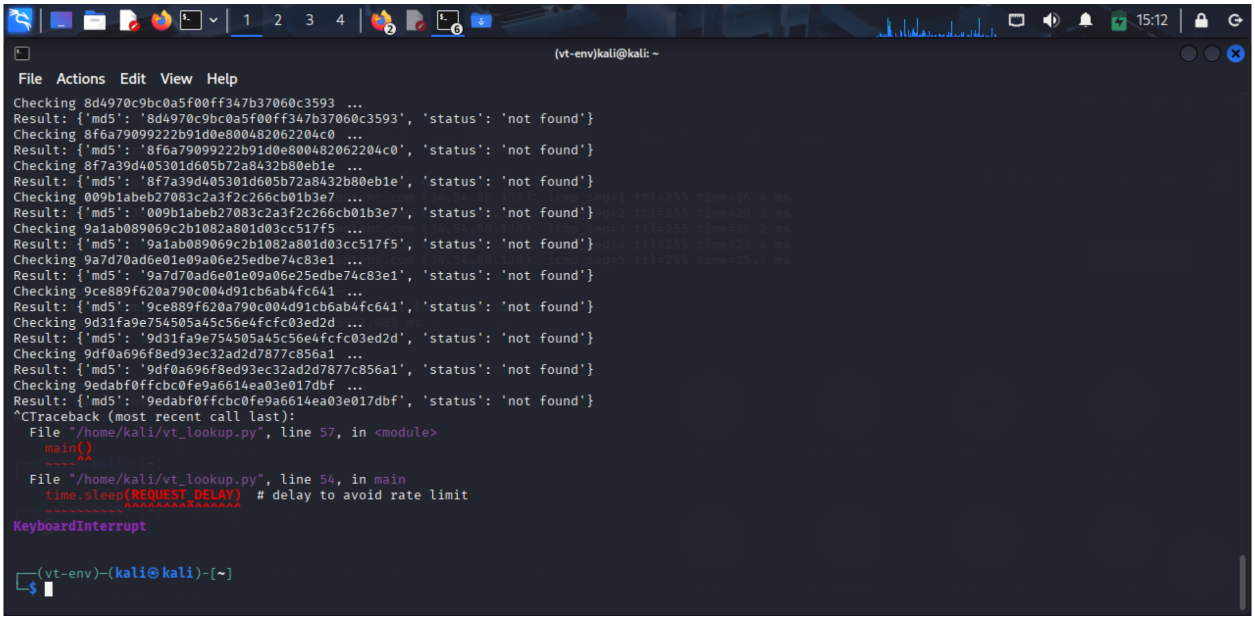
File Actions Edit View Help

```
┌──(kali㉿kali)-[~]
└─$ ping www.virustotal.com
nslookup www.virustotal.com

PING www.virustotal.com (34.54.88.138) 56(84) bytes of data.
64 bytes from 138.88.54.34.bc.googleusercontent.com (34.54.88.138): icmp_seq=1 ttl=255 time=16.4 ms
64 bytes from 138.88.54.34.bc.googleusercontent.com (34.54.88.138): icmp_seq=2 ttl=255 time=20.3 ms
64 bytes from 138.88.54.34.bc.googleusercontent.com (34.54.88.138): icmp_seq=3 ttl=255 time=16.2 ms
64 bytes from 138.88.54.34.bc.googleusercontent.com (34.54.88.138): icmp_seq=4 ttl=255 time=23.4 ms
64 bytes from 138.88.54.34.bc.googleusercontent.com (34.54.88.138): icmp_seq=5 ttl=255 time=15.7 ms
^C
--- www.virustotal.com ping statistics ---
6 packets transmitted, 5 received, 16.6667% packet loss, time 5016ms
rtt min/avg/max/mdev = 15.696/18.383/23.359/2.983 ms
Server:         10.0.2.3
Address:        10.0.2.3#53

Non-authoritative answer:
Name:   www.virustotal.com
Address: 34.54.88.138

┌──(kali㉿kali)-[~]
└─$

┌──(kali㉿kali)-[~]
└─$
```

(vt-env)kali@kali: ~

File Actions Edit View Help

```
Result: {'md5': '5c3c92c4887934a756d9de44e297bde9', 'status': 'not found'}
Checking 5d4bc22291c6accb2c338bfeffebfaee ...
Result: {'md5': '5d4bc22291c6accb2c338bfeffebfaee', 'status': 'not found'}
Checking 5faacfce2fb94f1812f865f9a025c662 ...
Result: {'md5': '5faacfce2fb94f1812f865f9a025c662', 'status': 'not found'}
Checking 6a2672821eff43df4eae4cd203814893 ...
Result: {'md5': '6a2672821eff43df4eae4cd203814893', 'status': 'not found'}
Checking 6a6908ec28ceb190abbdfb0c7fbe71d5 ...
Result: {'md5': '6a6908ec28ceb190abbdfb0c7fbe71d5', 'status': 'not found'}
Checking 6a8fde0e7c6e3572d925f23e15c84003 ...
Result: {'md5': '6a8fde0e7c6e3572d925f23e15c84003', 'status': 'not found'}
Checking 6add3e82067f0e6a11f8246d9684f014 ...
Result: {'md5': '6add3e82067f0e6a11f8246d9684f014', 'status': 'not found'}
Checking 6b776296e448d9e95eb55c149cf2ffca ...
Result: {'md5': '6b776296e448d9e95eb55c149cf2ffca', 'status': 'not found'}
Checking 6c3431693e67fc4ffa6012f0a25e9a5b ...
Result: {'md5': '6c3431693e67fc4ffa6012f0a25e9a5b', 'status': 'not found'}
Checking 6d0255d71b2c517ccc6dbfbad25f7044 ...
Result: {'md5': '6d0255d71b2c517ccc6dbfbad25f7044', 'status': 'not found'}
Checking 6fc078c584588e90a2f57fa93dcc29e8 ...
Result: {'md5': '6fc078c584588e90a2f57fa93dcc29e8', 'status': 'not found'}
Checking 6ff2f429c03cf0c05e04000561b72a02 ...
Result: {'md5': '6ff2f429c03cf0c05e04000561b72a02', 'status': 'not found'}
Checking 7a7253c1c3efcbc74b689858f2e72e98 ...
Result: {'md5': '7a7253c1c3efcbc74b689858f2e72e98', 'status': 'not found'}
Checking 7ac4b7abb5431c32283fc54fbe0b1b8d ...
Result: {'md5': '7ac4b7abb5431c32283fc54fbe0b1b8d', 'status': 'not found'}
Checking 7aeadfc4aff3bca4b8b53e545136e576 ...
Result: {'md5': '7aeadfc4aff3bca4b8b53e545136e576', 'status': 'not found'}
Checking 7bbbab38fc5b614330b2707fcdf56964 ...
Result: {'md5': '7bbbab38fc5b614330b2707fcdf56964', 'status': 'not found'}
```



✅ Conclusion

This investigation successfully implemented a repeatable detection and enrichment pipeline using Splunk and Zeek logs. By focusing on unique file hashes from a suspect internal host, I isolated indicators potentially tied to malicious behavior.

The analysis has:

- Validated that Zeek `files.log` is a rich source for file-level visibility
- Demonstrated the ability to bridge SIEM with external threat intel platforms (VirusTotal)
- Highlighted opportunities for **automation**, **alerting**, and **log context enrichment** to further empower SOC operations

Next Steps:

- Review `vt_results.csv` for positive hits
- Investigate high-confidence malicious hashes in Splunk (pivot on `dest_ip`, `uid`, `mime_type`)
- Consider creating a threat-hunting dashboard or Splunk alert based on this workflow

---

🧩 Appendix

Sample Enrichment Query (Splunk)

spl

```
index=main sourcetype=zeek_files_log [ | inputlookup vt_results.csv | fields
md5 ]
| stats count by md5, src_ip, dst_ip, mime_type, total_bytes
```

This pivot identifies which systems interacted with known-malicious files, based on external enrichment.